
A 100-MS/S 10-BIT 9.8-ENOB SAR ADC

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List of Abbreviations

Abbreviation	Description
ADC	Analog-to- Digital converter
BGR	Band gap reference circuit
BW	Bandwidth
CDAC	Capacitive Digital-to- Analog converter
CMOS	Complementary Metal-Oxide-Semi conductor
CLK	Clock
C_H	Holding capacitor
C_s	Sampling capacitor
DAC	Digital-to- Analog converter
DFF	D-flip-flop
DNL	Differential Non-Linearity
ENOB	Effective Number of Bits
FFT	Fast Fourier Transform
FIN	Input frequency
f_s	Sampling frequency
INL	Integral Non-Linearity
LDO	Low dropout regulator
LSB	Least significant bit
MIM	Metal- Insulator- Metal
MOM	Metal - Oxide - Metal
MOS	Metal-Oxide-Semi conductor
MSB	Most significant bit

NFFT	Number of FFT points
NCYC	Number of cycles
OSR	Over sampling ratio
OTA	Operational Trans conductance amplifier
PDK	Process Design Kit
PPM	parts per million
PVT	Process - Voltage - Temperature
P_Q	Quantization Power
SAR	Successive Approximation Register
SFDR	Spurious-Free Dynamic Range
SNR	Signal-to-Noise Ratio
SNDR, SINAD	Signal-to-Noise and Distortion Ratio
S/H	Sample and Hold
TG	Transmission Gate
TSPC	True single-phase clock
T_s	Sampling Time
V_{CM}	Common mode Voltage
V_{FS}	Full scale voltage
V_{GS}	Voltage between the gate and the source of the MOS
V_{REF}	Reference Voltage
$V_{REF,p}$	Positive Reference Voltage
$V_{REF,n}$	Negative Reference Voltage
V_{th}	Threshold voltage of the transistor

Abstract

A low-power 10-bit 100 MSps SAR ADC based on charge redistribution was designed for a 45 nm CMOS technology. The proposed design uses an efficient SAR algorithm (monotonic switching procedure with constant common mode level) to reduce power consumption due to capacitor charging by %50 compared to a conventional design, as well as reducing the total capacitor area by half. Sampling switches were bootstrapped for increased linearity compared to simple transmission gates. Another feature of the low power design is a fully dynamic comparator which does not have a static current. This module is work to get acceptable SNR for the output signal of the ADC equivalent to 9.8 ENOB

Chapter 1

Introduction

Analog-to-digital conversion is everywhere around us. In all forms of electronic equipment, the analog-to-digital converter links our physical world to digital computing machines. This development has enabled the marvelous electronics functionality that has been introduced over the last 30 years, from mobile phone to internet, from medical imaging machines to hand-held television.

Analog signals are connected to physical quantities like length, weight, volume, or more appropriate for the electronics field: voltage, charge, current, and magnetic flux. Analog electronics circuits manipulate these quantities by means of operations such as amplification, attenuation, and modulation. Analog-only circuits can do a lot of signal processing in a cheap and well-established way. Many signal processing functions are so simple that analog processing serves the needs (audio amplification, filtering, radio, television, and first-generation mobile phone). In more complex situations, analog processing, however, lacks required functionality and digital signal processing offers crucial improvements. The most important advantages of digital processing over analog processing are a perfect storage of digitized signals, unlimited signal-to-noise ratio, the option to carry out complex calculations, and the possibility to adapt the algorithm of the calculation to changing circumstances. If an application wants to use these advantages, analog signals have to be converted with high quality into a digital format in an early stage of the processing chain. And at the end of the digital processing the conversion has to be carried out in the reverse direction. The digital-to-analog translates the outcome of the signal processing into signals that can be rendered into a picture or sound. This makes analog-to-digital conversion a crucial element in the chain between the world of physical quantities and the rapidly increasing power of digital signal processing.

1.1 Basics in sampling Theory

1.1.1 Sampling

Signals are continuous in time. They have infinite time samples that need a memory of infinite size to store and process which is not practical. Fortunately, it is proven that any band limited continuous signal can be represented by finite samples with, theoretically, no loss in information. This was introduced by Harry Nyquist in his Sampling Theory.

Sampling theory

If a signal is band limited to BW hertz, it is fully represented by a number of points spaced by sampling time $T_s = 1/2BW$. Using this theory, we can discretize the continuous signal in time. ADCs use sampling as the 1st block to convert the analog input signal in time into a discretized one which exist only at certain time instant. The rate by which the sampling block at the ADC samples the input signal is called the Sampling Rate which is one of the most important metrics for the ADC and it plays a major role in determining the proper topology to Design an ADC.

Quantization

So far, the signal is considered analog of course it is existing at certain time instants due to sampling but, it can take any value which means that its amplitude is continuous so, we need to discretize the amplitude of the signal also such that our signal amplitude could take only certain values (levels) this process is called quantization and it is also an important metrics for the ADC and it plays a major role in determining the proper topology to Design an ADC.

Quantization noise

Analog-to-digital conversion is based on the concepts of sampling and quantization. Sampling is the process of capturing values, or samples, of the input signal at discrete points in time while quantization is the approximation of these values with discrete levels. The two processes are illustrated in fig 1.1 where the analog signal, which is continuous in both time and amplitude, is converted into a digital signal which is discrete in both time and amplitude. The accuracy of the quantization depends on the resolution, i.e. how many quantization levels there are. The resolution is normally expressed in number of bits n , which relates to the number of quantization levels R as $2^n = R$. The quantization errors can often be seen as white noise with the power calculated as in equation 1.1, where Δ is the step size between available quantization levels and X_{FS} is the full scale-swing.

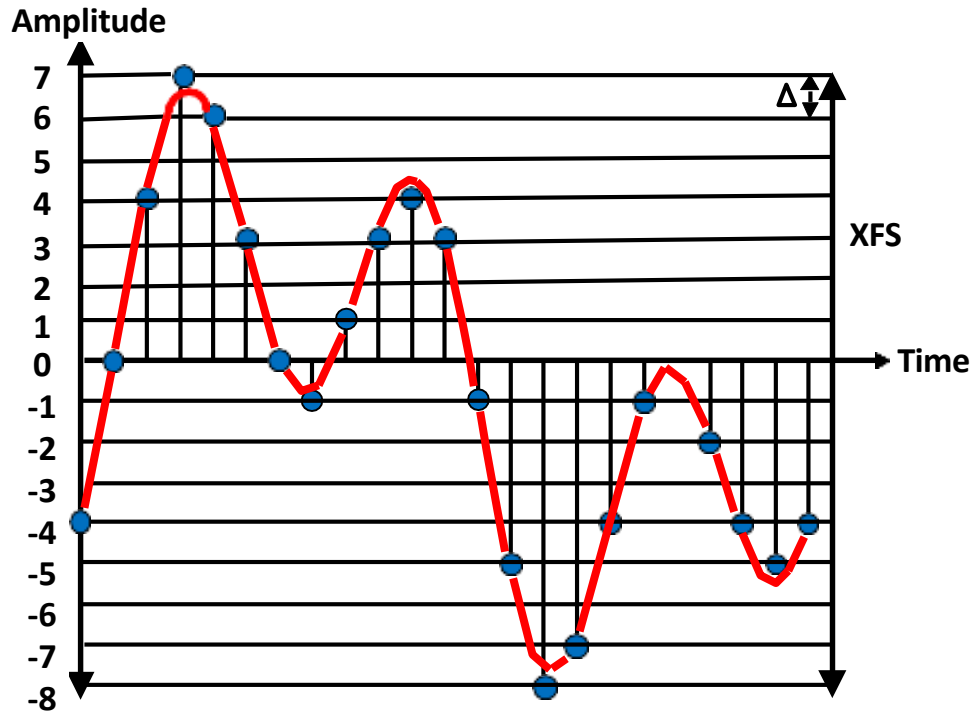


Figure 1 : Illustration of sampling and quantization - basic concepts of analog-to-digital conversion.

1.2 Required Specifications

The ADC in this work is a low power medium resolution analog to digital converter. The specifications of the ADC were provided by PDF related to cadence design. Table 1.1 shows the required specifications.

Table 1.1: ADC Specification

Parameter	Constrain	Min	Typical	Max	Unit
Technology	45nm				
Resolution	10				bits
ENOB	9.8				bits
Power	< 0.4				mw
Temperature	-30	27	150	$^{\circ}\text{C}$	
full scale	1	1.2	1.4	Volt	
T_{sampling}	10				ns

1.3 Architecture Selection

Architecture selection can begin with regards to the resolution, input bandwidth specified above. Since low-power operation is required in this design, architecture selection will be driven by an examination of the power consumption of previous ADCs. Fig 1.2 plots the power consumption of existing designs against the resolution. Here for 8-bit resolution the SAR ADC is optimum choice which consume low power (less than 10 mwatt). Also, the Sigma-Delta ADCs is a good choice. Figure 1.3 plots the SNDR with the signal BW which explains that to obtain high SNDR (equivalent to 10 ENOB) the proper topology is either SAR ADC or Sigma-Delta.

From the above discussion I choose to work on SAR ADC and my topology is SAR ADC to take the benefit of low power from SAR to achieve my specifications.

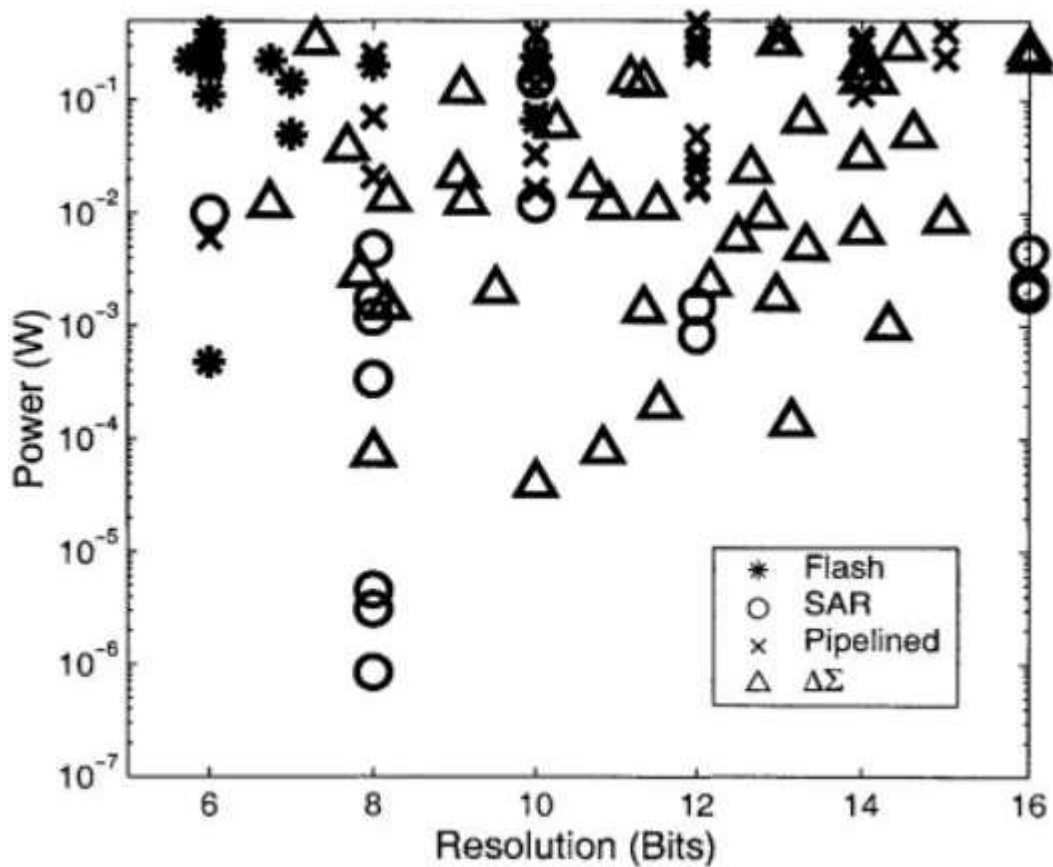


Figure 2 : Existing designs in a power-resolution space (data courtesy B. Ginsburg, MIT)

1.4 Thesis Organizations

The thesis consists of ten chapters. Chapter 1 is an introduction. Chapter 2 Presents the Modelling of the system. Chapter 3 Presents the DAC design, Chapter 4 discusses the Switches Imperfections and contain a Bootstrapped switch design. Chapter 5 contain Comparator topology and it enhancements. 6 Presents the SAR logic. 7 Presents the LDO Design.

Chapter 2

System design and modeling

A high-level model for the system was created. The high-level model consists of ideal sub-block models which are described in Verilog-A code. Although the sub-blocks do not take non-idealities into account, the overall functionality can be analyzed using these models. The models enable a systematic investigation of how each sub-block affects the performance of the ADC, since each sub-block can be changed from a high-level model to a transistor level implementation.

2.1 Capacitive DAC

In a SAR-ADC, the DAC is considered one of the most important blocks that determine the accuracy and conversion speed of the converter that also consumes a big portion of the overall power of the SAR-ADC. The DAC required in the SAR-ADC can be realized in various ways; e.g., capacitor-based DAC [1], switched-current DAC [2] or R-2R ladder DAC [3]. Among these architectures, the capacitor-based DAC has become the most popular because of its zero-static power consumption. Furthermore, in most technologies resistor mismatch and tolerance are greater than capacitor mismatch and tolerance.

A differential 8-bit capacitive DAC was implemented using Verilog-A. The capacitor array of the DAC is used to sample the input voltage during the sampling phase through switch. During the conversion phase, the bootstrap switch is turned off, and the output voltage of the DAC is determined based on the control signals of the SAR logic block.

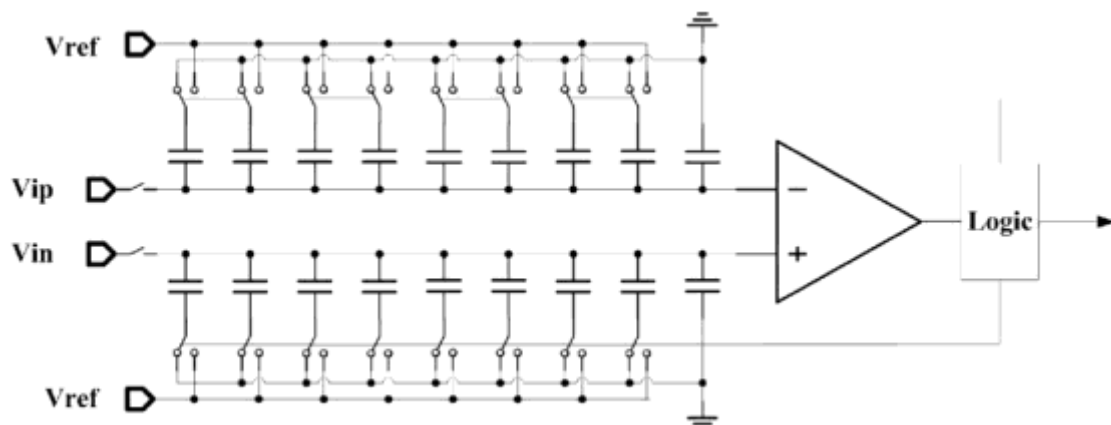


Figure 3 : Charge redistribution SAR ADC in ideal model

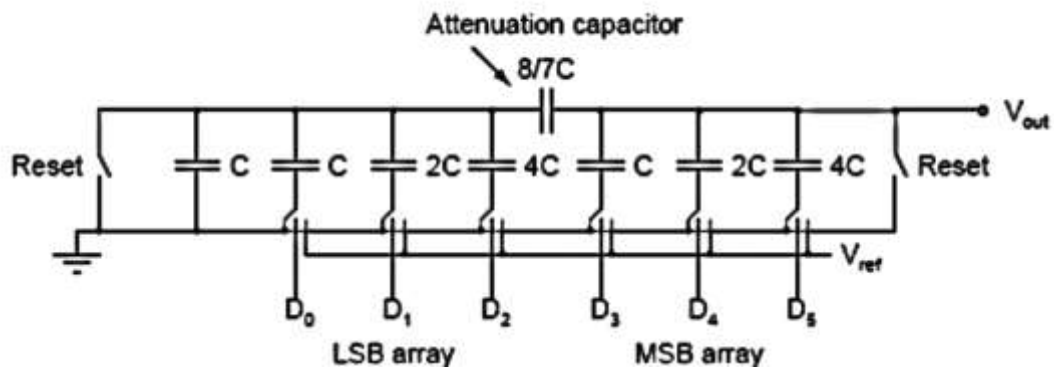


Figure 4 : Example of a capacitor array with a split-capacitor in Practical Model

2.2 Switching scheme

In the conventional switching scheme, the majority of the total energy dissipation is due to repeated charging and discharging of the large capacitor array. Several new techniques using alternative switching algorithms have been proposed which dramatically decrease this power consumption. The split-capacitor [4] and energy-saving [5] Techniques propose reduction in switching energy of 37% and 56% respectively, while the monotonic switching technique [6] offer savings of up to 81%. For the first glance it may seem that these techniques are outstanding but each technique has its disadvantages. First, is that the attenuation capacitor size is a non-integer multiple of the unit capacitance (Figure 4) which causes a large mismatch. Second, parasitic capacitances on sides of the capacitor array not only causes a gain error it also causes non-linearity. A Common Mode-Voltage (VCM) based switching scheme needs a third common mode reference which increases the design complexity. And the power saved by the switching is comparable to the power dissipated in the voltage reference circuit. The monotonic switching technique main disadvantage is that its common mode voltage is varying (Figure 5), the common mode voltage is approaching to ground. This variation puts a great complexity on the comparator design.

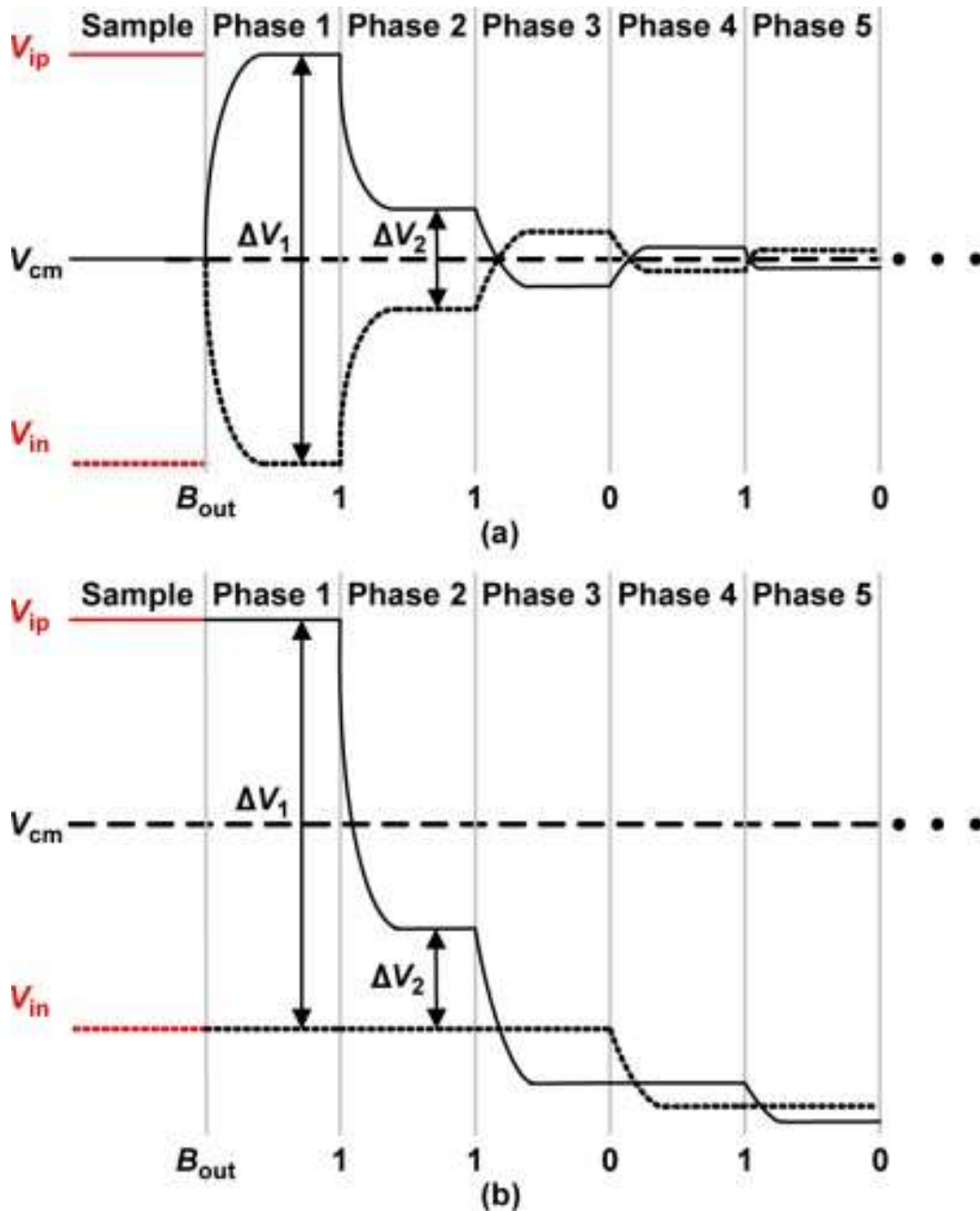


Figure 5 : Waveform of conventional switching (a) and monotonic switching (b) procedures

2.3 SAR logic

The SAR Logic Verilog-A code in appendix A.2 is describing a sequential algorithm consisting of ‘n+1’ states where ‘n’ is the DAC resolution.

2.4 Comparator

Verilog-A model of a simple differential comparator is generated as shown in appendix A.3. The comparator implemented is a positive edge triggered comparator that simply compares the two input voltages and generates the corresponding differential output.

2.5 Clocking

A 100MSps implies a conversion time of one complete cycle to be 10 nS. Timing is designed such that 1.66nS is used for sampling, 8.4nS are assigned for the quantization phase .During the 8.4nS of the quantization phase 10 comparisons are performed by the comparator In the quantization phase, the comparator is required to perform 10 comparisons, The comparator reset phase duration should be sufficient for SAR logic to generate the control signals and capacitive DAC to settle after switching, the comparison phase should be sufficient for the comparator digital output to be produced and stored in the comparator latch.

2.6 Operation verification

The circuit was simulated using the test bench shown in Figure 6 where input signal, clock signal and power supply all were generated by ideal voltage sources. The measurements were made with a sine wave as input signal and the ADC operating at a conversion speed of 100 MSps as stated in the specification. In order to analyze the digital ADC output, an ideal DAC was put converting the digital output to an analog signal. The fast Fourier transform (FFT) of this analog version of the output generates a spectrum denoting important metrics of the system such as ENOB, SNDR and SFDR.

Figure 7 shows the input and output of the ADC, after converting the digital output to an analog signal. The input frequency is $(503/1024) \cdot (100 \text{ MHz})$. Figure 8 and Figure 9 presenting the spectrum of the output signals achieving an ENOB of 10 bits successfully indicating that the system is functionally correct

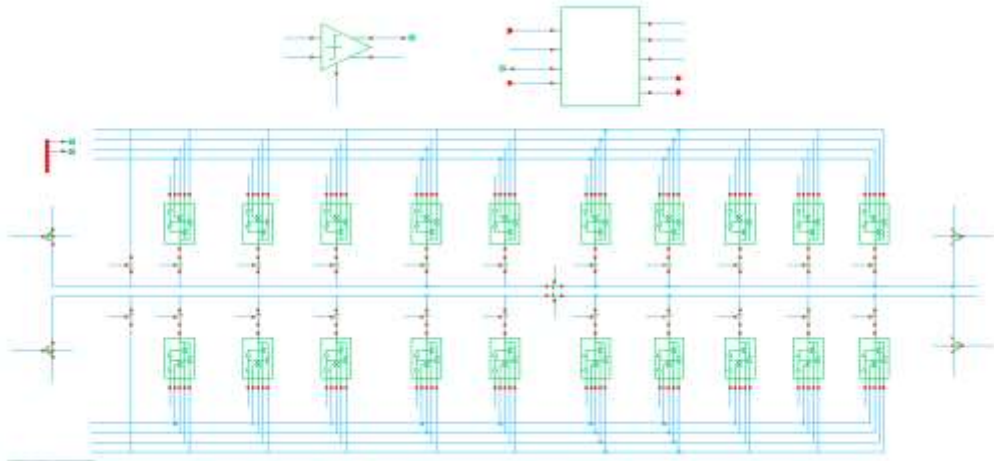


Figure 6 : Ideal Model of SAR ADC

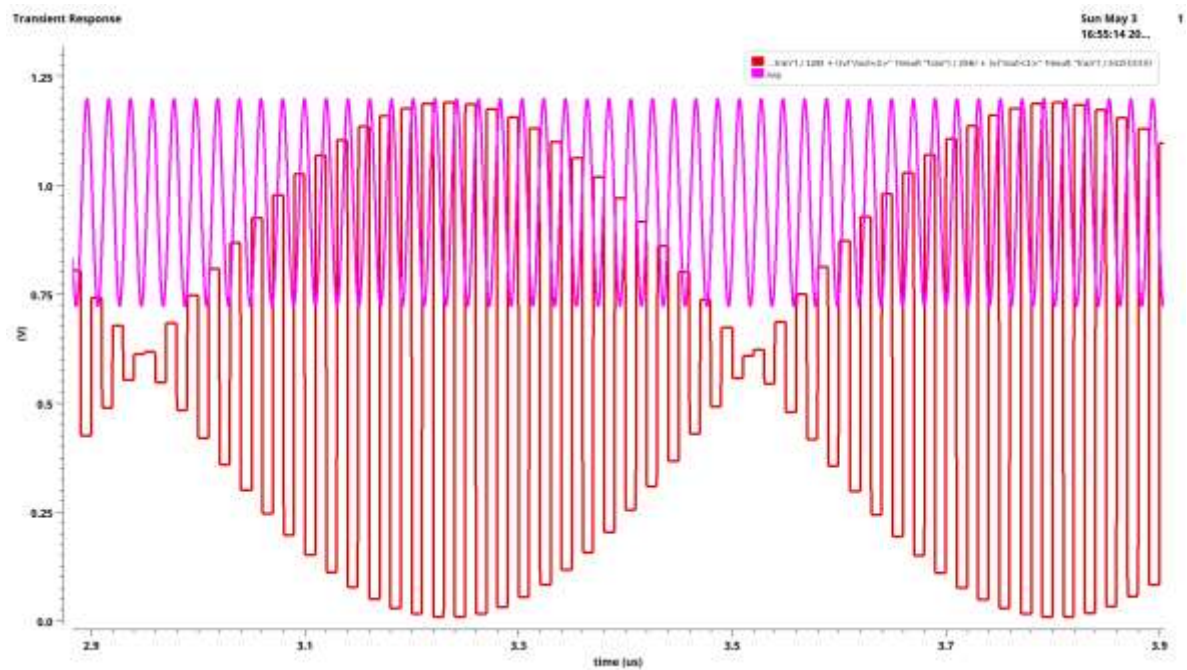


Figure 7 : Ideal output of SAR ADC

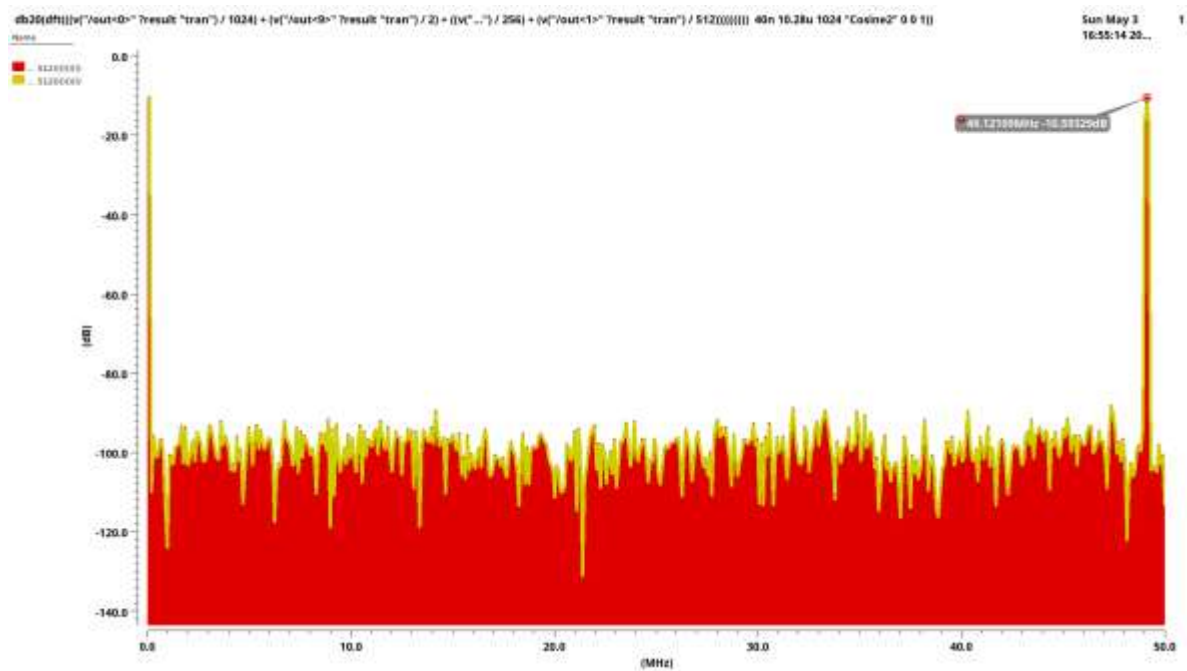


Figure 8 : Spectrum of SAR ADC at input frequency

Chapter 3

Capacitive DAC

Specifications and design requirements of the DAC are stated in Table 3.1

Table 3.1: Specifications and design requirements of the DAC

Parameter	min	typ	max	unit
Temperature	-30	27	150	$^{\circ}C$
DNL	-0.5	-	0.5	LSB
INL	-0.5	-	0.5	LSB
ENOB		10		bits

3.1 Type of capacitors used

Three kinds of capacitors are commonly used in IC applications, MOS capacitor, metal–insulator–metal (MIM) capacitor, and metal–oxide–metal (MOM) capacitor. A great caution was taken while choosing the most suitable type of capacitors for the CDAC, because the system performance is greatly affected by the CDAC performance. The CDAC must maintain good characteristics across PVT and corners while also achieving the best area possible to reduce the fabrication cost. MOS capacitor has the highest capacitance density per unit area among others. However, due to the disadvantages of non-linearity, high temperature coefficient, low break-down voltage, sensitivity to process variations, and gate-tunneling, it's not a good suite for our application. Fortunately, with the dimension shrinkage, the capacitance densities of MIM and MOM capacitors are increasing and reaching the density of MOS capacitors which makes them perfect replacements for MOS capacitors as they don't suffer its disadvantages. MIM and MOM capacitors share many good qualities such as low parasitic capacitance, good matching characteristics and high capacitance density (up to $2\text{fF}/\mu\text{m}^2$). However, MOM is gaining popularity over MIM as its density further increases with tech scaling and, the most important advantage over MIM, it requires no additional fabrication masks or process steps. MOM capacitor is realized through the metal interconnections, as shown in Figure 9. While MIM capacitor uses a different dielectric material (ϵ_x) to enhance its capacitance density, as shown in Figure 10.

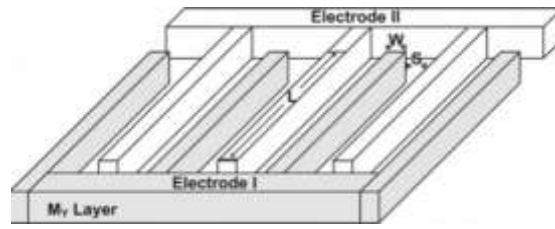


Figure 9 : MOM capacitor structure layout

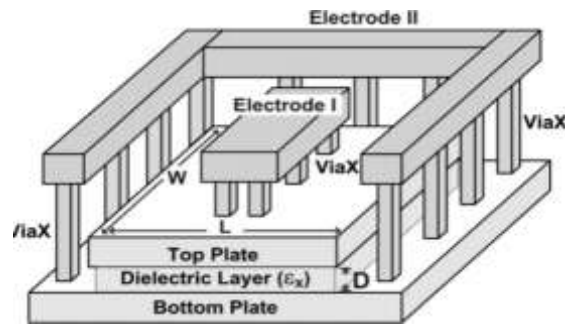


Figure 10 : MIM capacitor structure layout

3.2 Unit Capacitance calculation

The impact of thermal noise can be calculated as follows:

- The quantization noise power with $V_{FS} = 2 * V_{ref} = 2.4V$, due to the differential architecture, and $n=10$ bits $PQ = \frac{\delta^2}{12} = \frac{(2/2^{10})^2}{12} =$
- Taking the thermal noise to be much smaller than the quantization noise, 10dB lower, gives a lower limit on the total capacitance.
 $(KT/C_{total}) < (1 * Pq/10)$ thus $C_{total} >$

This small lower limit on the total capacitance indicates that the thermal noise is an order of magnitude smaller than the quantization noise even with the smallest available unit capacitance in the PDK. Therefore, the unit capacitance is determined based on the allowable capacitor mismatch.

Capacitor mismatch

Mismatch between unit capacitors can cause deviation from the ideal transfer curve of the DAC, resulting in a distorted output, degrading the SNR of the ADC.

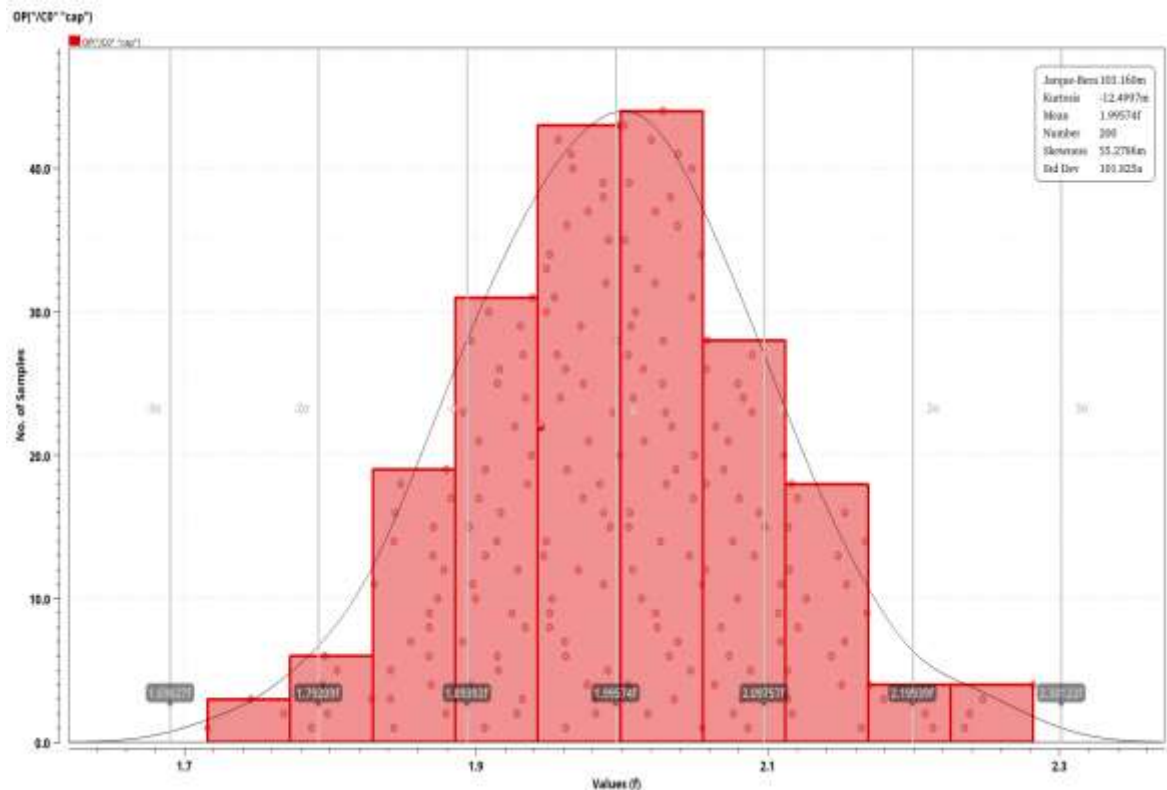


Figure 11 : Mismatch simulation for the smallest capacitor provided by the PDK

This makes matching of unit capacitors critical for the accuracy and resolution of the entire ADC.

For an n-bit resolution DAC, the standard deviation of a unit capacitor mismatch ratio, (σ_u) must satisfy $\sigma_u < 1 / (6 \sqrt{2^{(n-2)}})$ to have $DNL_{max} < 1/2$ LSB with 99.7% confidence level. Thus, for an 10-bit DAC $\sigma_u < 1 / (6 \sqrt{2^{(10-2)}})$

Figure 11 shows Monte Carlo simulation for the smallest capacitor provided by the PDK. The simulation is held for 200 samples, and the standard deviation for the minimum size capacitor is determined to be $\sigma_u = 1.99574 \times 10^{-3}$ which is far less than the previous specified value indicating that we can use the minimum size capacitor without any problems. Although, the unit capacitance is chosen to be 15fF.

3.3 DAC Switches

The switches used in the design are basically Muxes

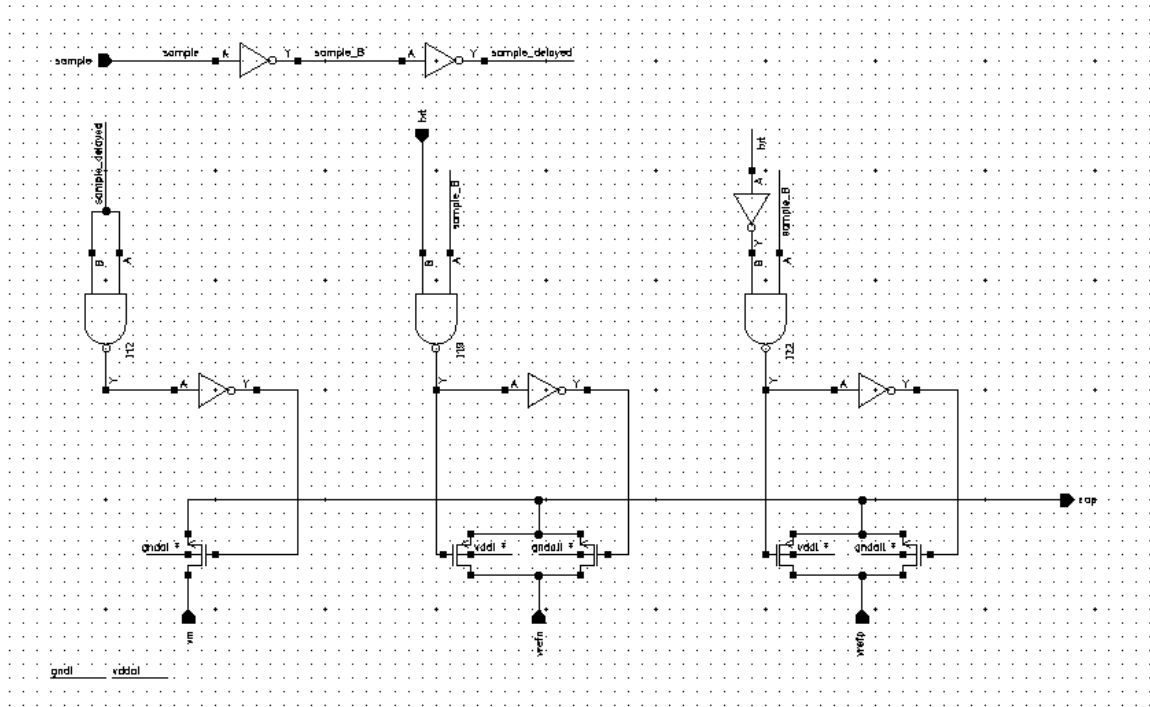


Figure 12 : schematic view of the used switch

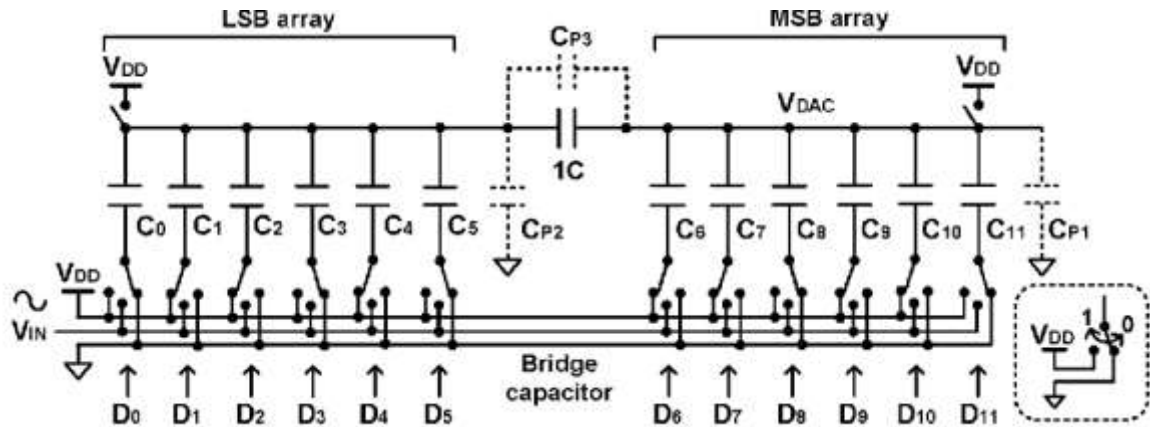


Figure 13 : Schematic view of the CADAC

3.4 Simulation Results and Verification

Table 3.2: Achieved Specifications

Spec	Achieved
ENOB	9
DNL	LSB
INL	LSB

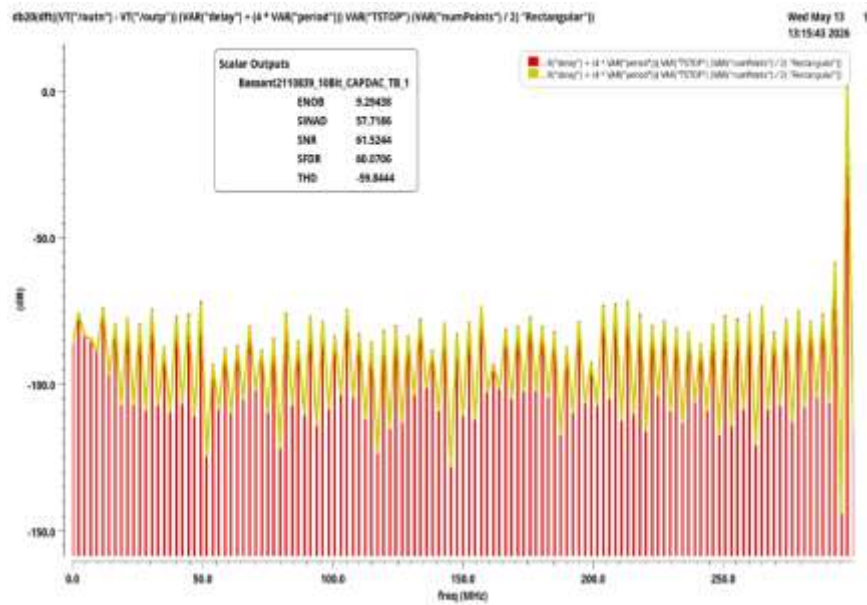


Figure 14 : ENOB Parameter to express the performance of DAC

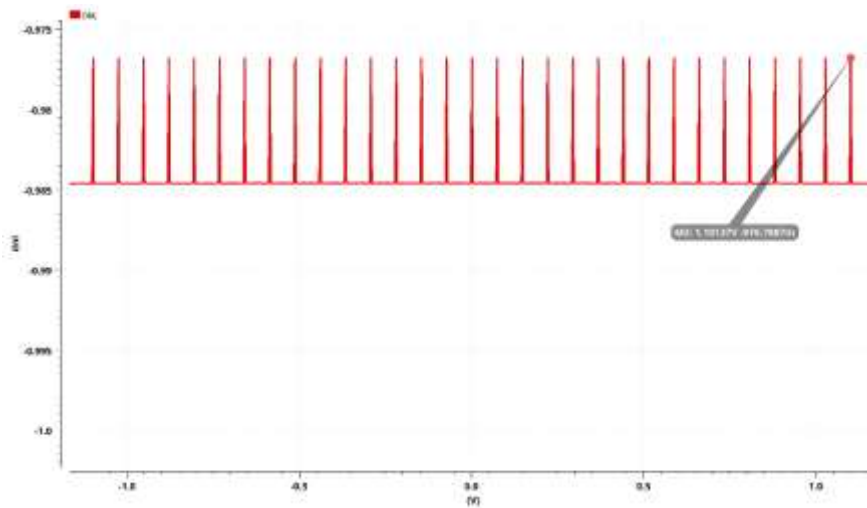


Figure 16: Differential Non-Linearity

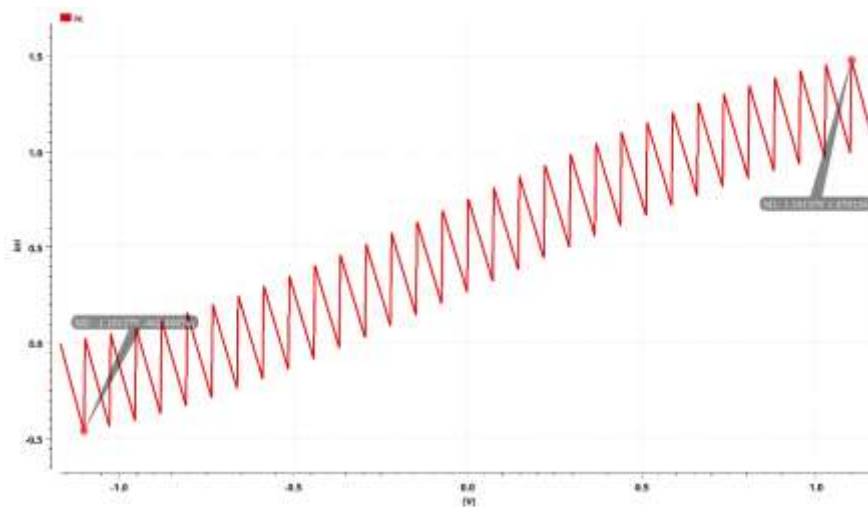


Figure 15: Integral Non-Linearity

Chapter 4

Sampling Switch

Sampling of the time-varying input signal is the first step in any type of Analog to Digital (A/D) conversion. For high-resolution and high-speed A/D converter, a high performance Sample and Hold (S/H) circuit is needed as its front-end component. Linearity of the front-end Sample and Hold circuit directly impacts the linearity of the consequent stages of A/D converter. The main reason of non-linearity in sampling switch is the variations of ON resistance with input signal which produce harmonic distortion when sampling high-frequency signals. This limits the SINAD and the ENOB to be obtained from the sampling switch.

4.1 Sources of non-linearity in sampling switch

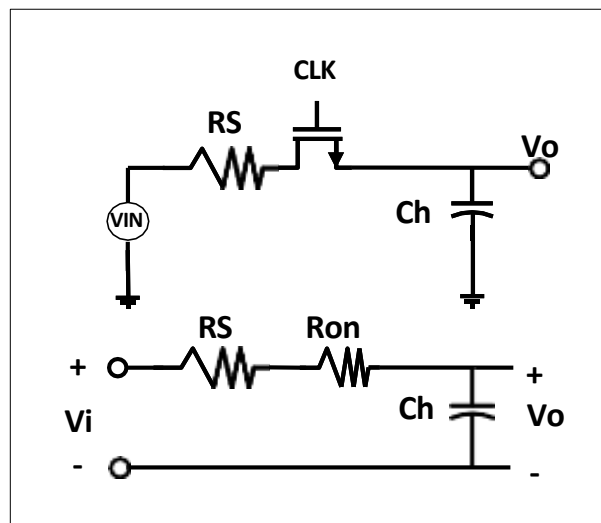


Figure 17 : Simple Track and Hold Circuit

Fig 17 shows the simplest track-and-hold circuit. When CLK is high, the switch is turned on and charge is stored on the capacitor to V_{IN} . When CLK is low, the switch turns off and the capacitor will hold the sampled voltage.

The on-resistance of this switch is given by:

$$R_{on} = 1 / (\mu_n \times C_{ox} (W / L) (V_{gs} - V_{TH}))$$

Equation shows the the dependence of on-resistance on V_{IN} in two terms:

- I. V_{gs} , as the input is applied to the source terminal.
- II. V_{TH} , as it depends on the voltage between source and bulk V_{SB} which vary with the input signal.

This dependency is the cause of the non-linearity and harmonic distortion in the sampling switch so, we need constant R_{on} for better ENOB.

The on-resistance R_{on} together with the sampling capacitor C_s form the RC time-constant that defines the -3 dB bandwidth during track mode. It is defined as follows:

$$f_{3dB} = (1/2 \times \pi R_{on} C_s) = (\mu n \times C_{ox} W/L (V_{gs} - V_{TH}) / (2 \times \pi C_s))$$

So, to achieve high input bandwidth for a fixed sampling capacitor the on-resistance needs to be small this achieved by decreasing L (technology scaling), increasing W and increasing the overdrive voltage V_{ov} . The last two options reduce the accuracy of the sampling switch through increasing the charge injection which is another source of non-linearity in sampling switch. It is defined as follows:

$$V_{pedstal} = (Q_{gate}/2/C_s) = (WLC_{ox} (V_{DD} - V_{in} - V_{TH})/2)/C_s$$

Where, $V_{pedstal}$ is a decrease in the hold voltage due to charge injected from the sampling switch when it turns from on to off. It is obvious from the above relation that it depends on the input voltage through V_{gs} and V_{TH} exactly like R_{on} .

Also, there is a lot of Problems in NMOS Switch but I will use it because it has the lowest power and making a great enhancement.

4.2 Design Strategy

When designing the NMOS switch, it is the most critical switch M_S which has to be large enough to ensure low resistance but not too large considering power dissipation and charge-injection effects. The width of M_{SS} was chosen to () μm after tests to achieve the required R_{on} and with minimum power consumption. At this value $R_{on} = 115 \text{ ohm}$ and this result is obvious from fig 18 that for different values of V_{IN} the on resistance is approximately constant and equal 115 ohm so, the acquisition time needed by the input to track the output with N bit accuracy is given by:

$$t_{acq} = 0.69 R_{on} C_{Hold}$$

That defines a maximum on resistance for the core switch that is given by:

$$R_{on} < (T_{track} / 0.69 R_{on} C_{hold})$$

$T_{track} = 1.67ns$ so R_{on} must be less than $126ohm$ and it is achieved in our design as $R_{on} = 115 ohm$

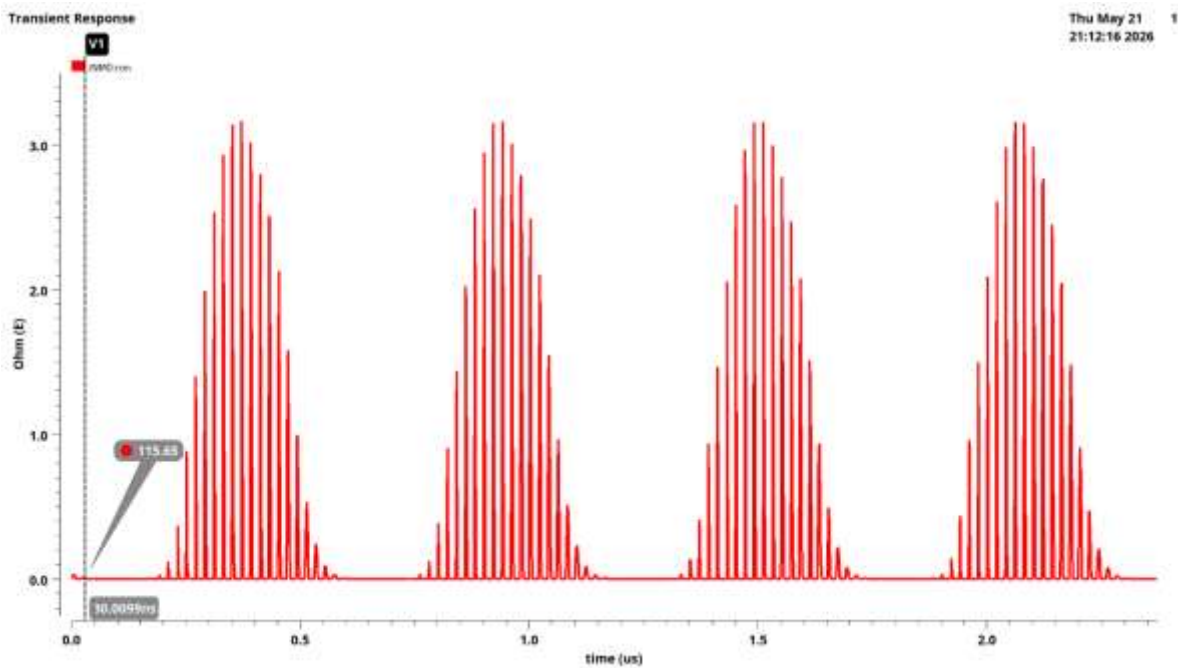


Figure 18 : RON for the NMOS Sampling Switch

Table 4.1: Track and Hold Specifications

Specifications		Required		
ENOB		> 10 bits		
Sampling clock frequency		100 MHZ		
T_{on}		1.67n		
VDD	1 V	1.2 V	1.4 V	
Temperature	-30 C ⁰	27 C ⁰	150 C ⁰	
C_{Hold}		1.92pf		

4.3 Simulation and Results

The test bench in fig 19 is used to measure ENOB, pedestal step and offset error for different design points. The 1st stage is the NMOS switch with $C_L = 1.92pF$ at its output equivalent to the total DAC capacitance, and the 2nd stage is ideal track and hold circuit to obtain sample and hold output to easily get the FFT, then we isolate between the stages using buffer.

Circuit gives an SINAD of 21.86 dB (3 bits) which is a clear improvement. we use the differential architecture as shown in Fig 19 which cancel the 2nd harmonic distortion and increase the SINAD to 44.08 (7 bits), and we have a pedestal step but as we work differential so, it has no effect on our system.

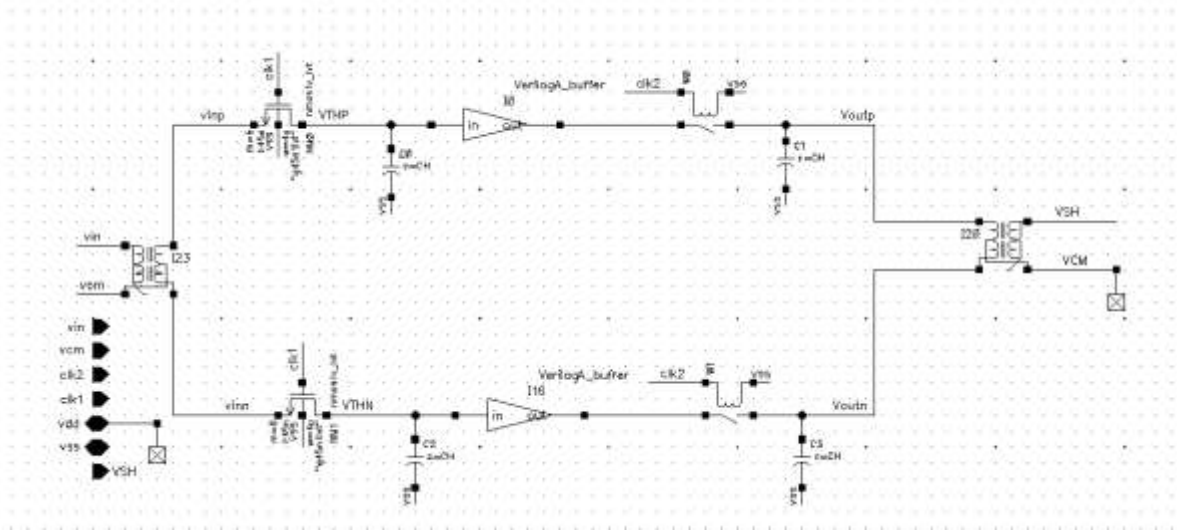


Figure 19 : Sampling switch test bench

Table 4.2: Track and Hold Simulation Parameters

Simulation	Transient
NFFT	2^{10}
NCYC	503
f_s	100MHZ
F_{in}	$(NCYC/NFFT)/T_s$
C_{Hold}	1.92p
VDD	1.2 V

4.4 Corner Simulations

Now, we make simulations across corners (ss, ff, fs, sf) each for temperature (27,-30, 150) each for supply voltage (1, 1.2, 1.4) and the result will be shown at [Table 4.3](#).

4.5 Achieved Specs

Table 4.4 summarize the achieved results from the designed NMOS switch which match our system requirements.

Table 4.3: ENOB simulation across corners for T=27,-30,150c and vdd=1, 1.2, 1.4V

Temperature = 27C ⁰	Temperature = -30C ⁰	Temperature = 150C ⁰
Supply voltage = 1	Supply voltage = 1	Supply voltage = 1
SS: ENOB = 2.79	SS: ENOB = 2.712	SS: ENOB = 2.721
FF: ENOB = 4.039	FF: ENOB = 3.945	FF: ENOB = 3.843
FS: ENOB = 3.722	FS: ENOB = 3.616	FS: ENOB = 3.57
SF: ENOB = 3.018	SF: ENOB = 2.934	SF: ENOB = 2.93
Supply voltage = 1.2	Supply voltage = 1.2	Supply voltage = 1.2
SS: ENOB = 2.79	SS: ENOB = 2.712	SS: ENOB = 2.721
FF: ENOB = 4.039	FF: ENOB = 3.945	FF: ENOB = 3.843
FS: ENOB = 3.722	FS: ENOB = 3.616	FS: ENOB = 3.57
SF: ENOB = 3.018	SF: ENOB = 2.934	SF: ENOB = 2.93
Supply voltage = 1.4	Supply voltage = 1.4	Supply voltage = 1.4
SS: ENOB = 2.79	SS: ENOB = 2.712	SS: ENOB = 2.721
FF: ENOB = 4.039	FF: ENOB = 3.945	FF: ENOB = 3.843
FS: ENOB = 3.722	FS: ENOB = 3.616	FS: ENOB = 3.57
SF: ENOB = 3.018	SF: ENOB = 2.934	SF: ENOB = 2.93

**Table 4.4:
and Hold**

Specification	Nominal	Units
ENOB	3	bits
power	65.35	n watt

Specifications

**Track
achieved**

Chapter 5

Comparator

In this section we will discuss the comparator block, it is one of the most important blocks in ADC, it is the block which will take the digital decision of conversion for each bit according to the differential input of the block, so it will make one of the output sides equal VDD and the other will be grounded, for example if $V_{in} > V_{th}$ therefore the output of the comparator will be $V_{OUTP} = V_{DD}$ and $V_{OUTN} = 0$ (GND). Sometimes we can use the comparator to detect the polarity of

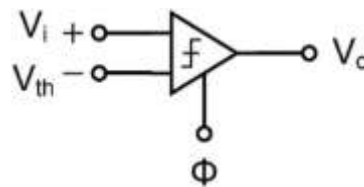


Figure 20 : simple comparator

The differential analog input, so the comparator will generate (1 or 0) accordingly. Or we can use it to compare a single ended input to a threshold or a reference voltage.

5.1 Comparator Requirements

There are many requirements for the comparator should be considered in the design:

1. Large amplification: As we need to translate a few mVs or uVs (differential input) to VDD or GND (“0” and “1”), so the comparator should have large gain (power of 10).
2. Wide bandwidth: The comparator should have high speed to make a quick right decision.
3. Accuracy: The accuracy of the comparator represented in the offset and the noise. The offset will have the effect of shifting the characteristics of the comparator, which means that instead of crossing the threshold voltage V_{TH} after shifting the characteristics by $\pm\Delta$, now crossing $V_{TH} + \Delta$ will determine the decision which should be taken. The offset voltage is constant for one comparator but it is different from one comparator to another one, but The noise is the varying offset for one comparator which means that the characteristics of one comparator will be shifted randomly

4. Low power consumption: It will consume large portion of the total available power of the system
5. No hysteresis: The comparator should not have any memory, which means that the previous comparator decision should not affect the following one.
6. No metastability: Metastability will happen when the comparator is not able to make a full-scale decision, which means that the output of the comparator will not be a strong "1" or a strong "0". It will be undefined decision which is worse than wrong decision.
7. Wide common mode input range: We need wide range of the threshold (reference) voltage.
8. High CMRR: CMRR is the ratio between differential gain over common mode gain, so it is required to have high differential gain and small common mode gain. As we need to make the decision according to the differential input only which is around mVs or uVs and the output is VDD or GND. And when AC common mode input is applied to both inputs of the input pair will not affect the decision of the comparator.

5.2 Imperfections of Comparator

1. Comparator offset voltage:
 we have two types of offset Static Offset which is random offset due to random mismatch of transistors and resistive loads, and it can be minimized by good design (maximize the area) and it is dominant in preamplifiers.
 Dynamic Offset Due to imbalanced switching effects and capacitive load mismatch which is dominant in latches and may cause nonlinear errors (for example: INL and DNL).
2. Comparator noise:
 Thermal noise: It can be modelled like the offset, but it changes for the same comparator, it can be reduced by increasing (gm) or decreasing (the bandwidth), but this will increase the power, decrease the speed.
 Kickback noise: It happens as a result of coupling between output and input. It can be reduced by decreasing the capacitance (but this will have a bad effect on mismatch) or decreasing the output swing. And may affect the decision of the comparator.

3. Hysteresis:

The hysteresis determines the resolution of the comparator. It happens as a result of imbalanced node voltages.

We can solve this problem by resetting these nodes each state. We can do overdrive recovery test to test that the comparator is able to resolve to the right output when small input ($\pm 0.5\text{LSB}$) is applied in a cycle after a full-scale input is applied.

4. Metastability:

The comparator will not generate a clear Zero or One after time T_s and Circuit will behave ambiguously. So we have a limit on the input.

5.3 Comparator topologies

1) Straight forward amplification at which multistage amplifiers could be used to get high gain and it will be a free running topology which means that there is no need for a clock to start the operation. The advantage of this topology is that the design is so simple and it is consisting of multistage so we can get high gain.

The disadvantages of this topology are, the design is relatively slow as we have high gain, so we will have a small bandwidth and as it is a free running topology, so it will work continuously and will consume large power.

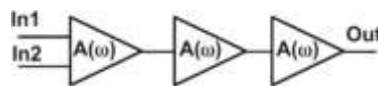


Figure 21 : straight forward amplification

2) Latched comparator in this topology there are two main stages, Preamplifier stage and Latch stage. This topology is clocked.

The preamplifier is used to suppress the mismatch and the offset of the latch and to isolate between the sensitive analog input and the regeneration section. It has low gain (around 10), so it allows high speed.

The latch is a Regeneration section, used to regenerate the output signal of the preamplifier to reach V_{DD} or GND according to the differential input. The Advantages of this topology is that it has less stability issues, less power consumption as it is a clocked topology (will not work continuously). This topology is the more common in designing comparator as it will give less power consumption and faster operation.

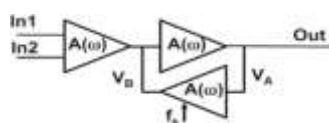


Figure 22 : Latch comparator

5.4 Architecture of latched comparator

In this section I will discuss an actual implementation of the comparator to know their advantage and disadvantage.

5.4.1 Strong-arm Latch

This architecture consists of Preamplifier M1 and M2, Latch consist of two cross coupled inverters M3: M6 and reset switches to prevent hysteresis S1: S4 as shown in fig 23

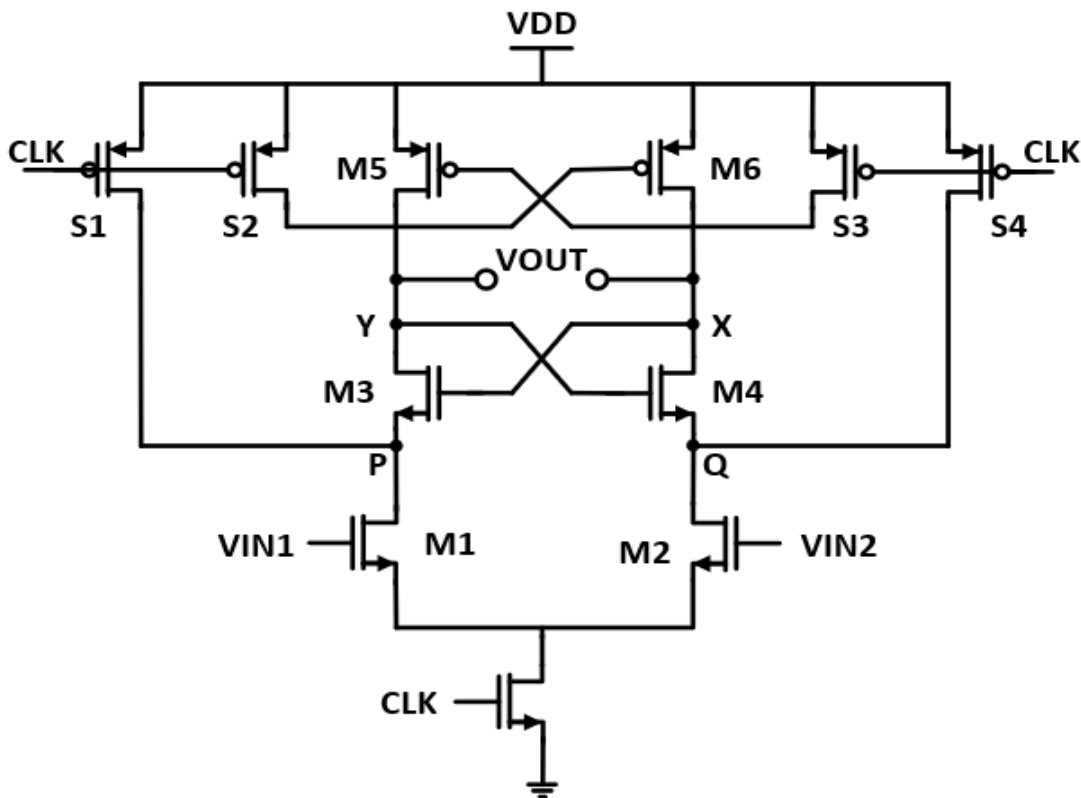


Figure 23 : Strong-arm architecture

It has two phases of operations (Precharge phase when CLK=0 and Amplification phase when CLK=1).

In Precharge phase (CLK=0): the reset switches will be ON and they will charge nodes (P, Q, X, Y) with VDD to prevent hysteresis. P & Q charged through S1, S4 and X, Y through S2, S3.

In Amplification phase (clk=1): input pair transistors will be ON and they begin to discharge node (P, Q) and the faster one is according to the differential input, then M3 and M4 will be ON when one of the nodes (P, Q) equal $V_{DD} - V_{THP}$ then nodes (X, Y) start to discharge and then the positive feedback (regeneration) will operate and helps one of the outputs to reach VDD and the other reach the GND.

This architecture has the advantage of dynamic operation in the pre amplification stage as when the CLK=0 the PMOS transistors will be ON and NMOS will be OFF and when CLK=1, the NMOS transistor will be ON and the PMOS transistors will be OFF, so there is no static current which give the architecture the advantage of low power consumption.

But it is suffering from kickback noise as the input and the output is directly coupled by the parasitic capacitance between them. Also, this architecture has large input referred offset as it does not have large gain.

5.5 Comparator Specifications

As shown in Table 5.1, from the required specifications, we decided to go through design a strong- Arm comparator and its enhancements as we have a tradeoff between powers, delay and offset.

Table 5.1: Comparator specifications

Spec	Value
Power	< 3 mV
Delay	< 500p
Load	50f
Input cap	< 200f
offset	< $0.5\Delta(LSB)$

5.6 Enhancement on the used Architecture

5.6.1 First enhancement

As shown in Table 5.1, we will find that the required input capacitance is 2f and it's required to derive 150f load which is too large by comparing it with the input capacitance. Then we decided to put a buffer to make the comparator capable to derive this load while keeping the input capacitance as it is, so a couple of inverters (Fig 24) was designed to behave as a buffer as shown in Fig 25.

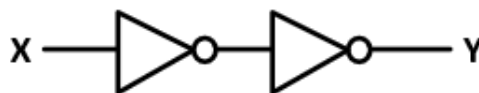


Figure 24 : Two cascaded inverters as a buffer

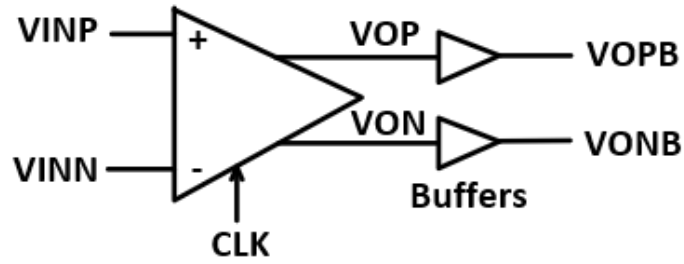


Figure 25 : Comparator followed by buffers

5.1.1 Second Enhancement

As we will use a clocked comparator, it will take the decision in the pre amplification or latch phase (assume in the first half cycle of the CLK period), and then will reset its internal nodes in the precharge phase or reset phase (assume in the second half cycle of the CLK period), so we want to isolate the next block of the system (SAR logic) from this phase while keeping the taken decision as it is, so we have to use SR latch to save the previous output (of the pre amplification and latch phase) for the next half cycle (of the reset phase).

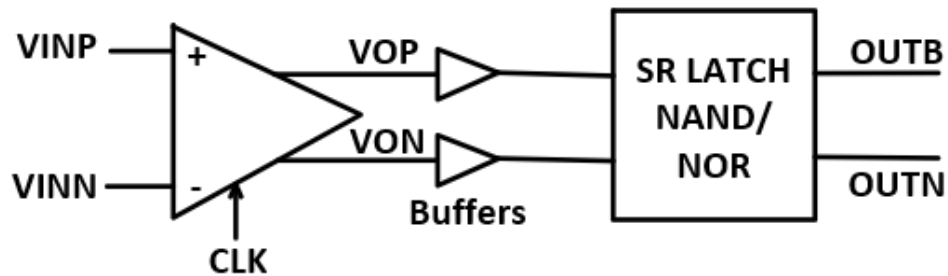


Figure 26 : Comparator followed by buffer then SR latch (NOR or NAND based)

5.7 Simulation Results

5.7.1 Strong Arm Comparator

I will show the simulation results.

1. Positive DC input to the comparator:

As shown in Fig 27, when a dc input (20mV) as a differential input to the comparator, as this means that V_{INP} which is the V_{GS} of the NMOS is greater than V_{INN} which is the V_{GS} of the NMOS, so the output of the differential pair V_{ON} will discharge faster than V_{OP} , and this is obvious in fig 27, $V_{ON} = 277\text{mv} < V_{OP} = 389\text{mv}$. Then after regeneration starts, the output of the comparator V_{OP} goes to VDD and V_{ON} to GND.

As shown Fig 28, the output of the buffer is smoother than the output of the comparator, V_{OPB} (buffered V_{OP} positive output of the comparator) = VDD and V_{ONB} (buffered V_{ON} negative output of the comparator) = GND.

As shown in Fig 29, the output of SR latch insensitive to what happens in the reset phase, OUTP (will be positive output of the comparator block in the system) = VDD and OUTN (will be negative output of the comparator block in the system) = GND.

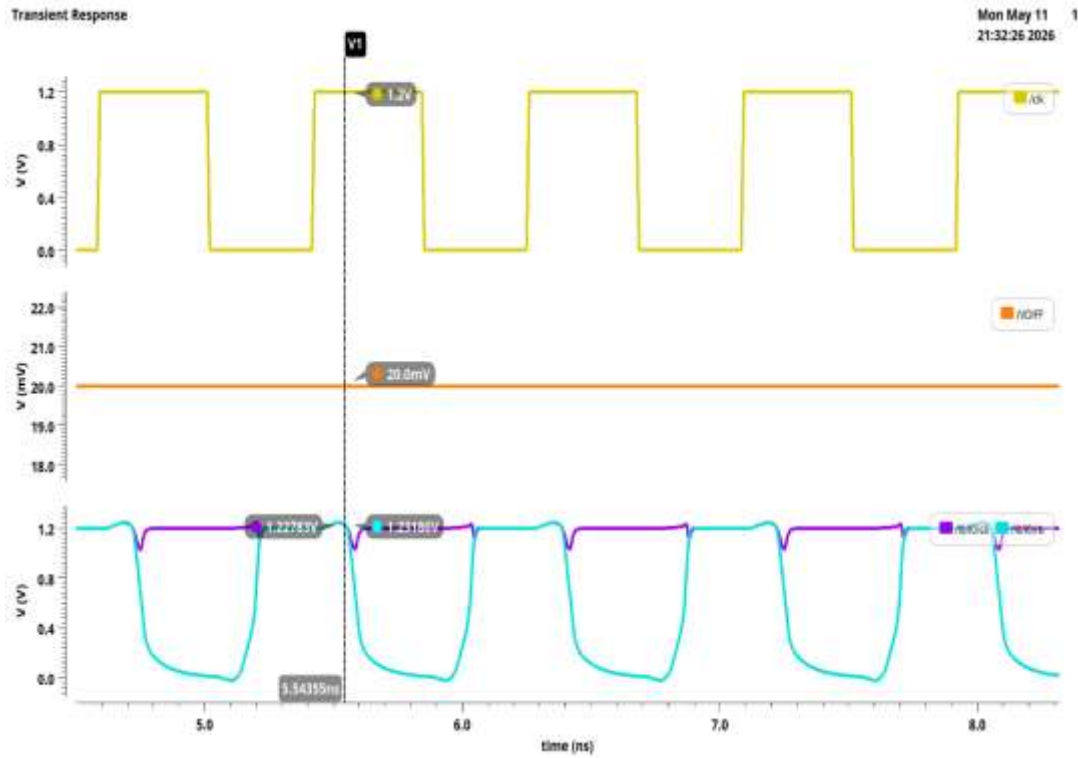


Figure 27: diff pair output and comparator output

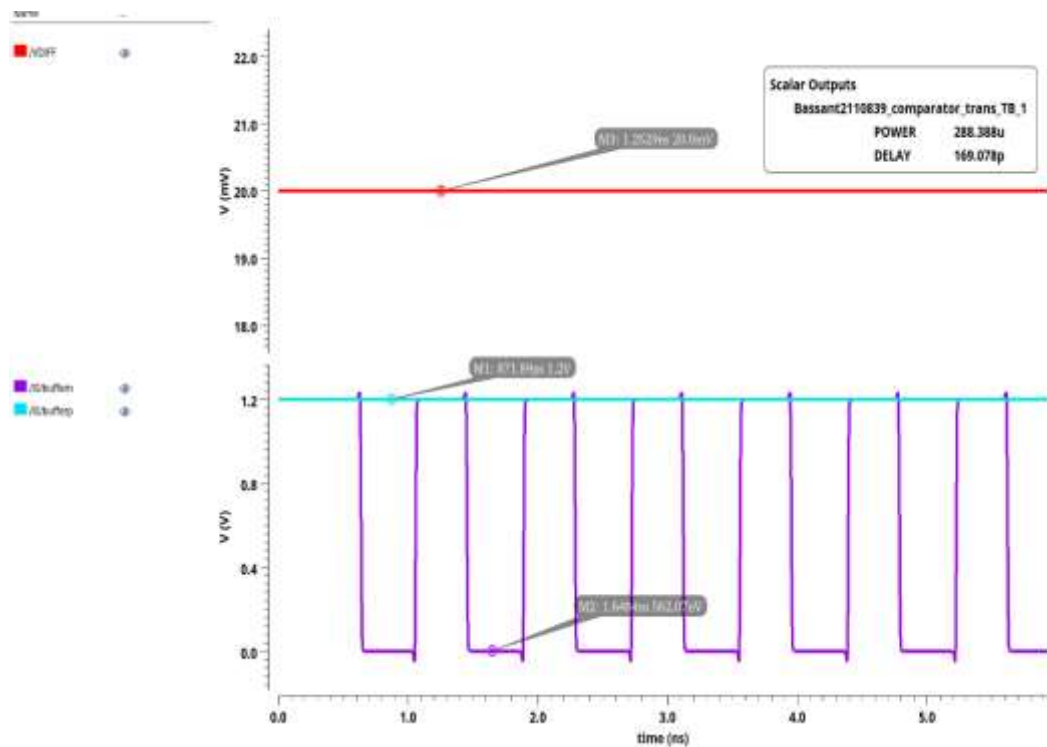


Figure 28: Comparator and buffer output

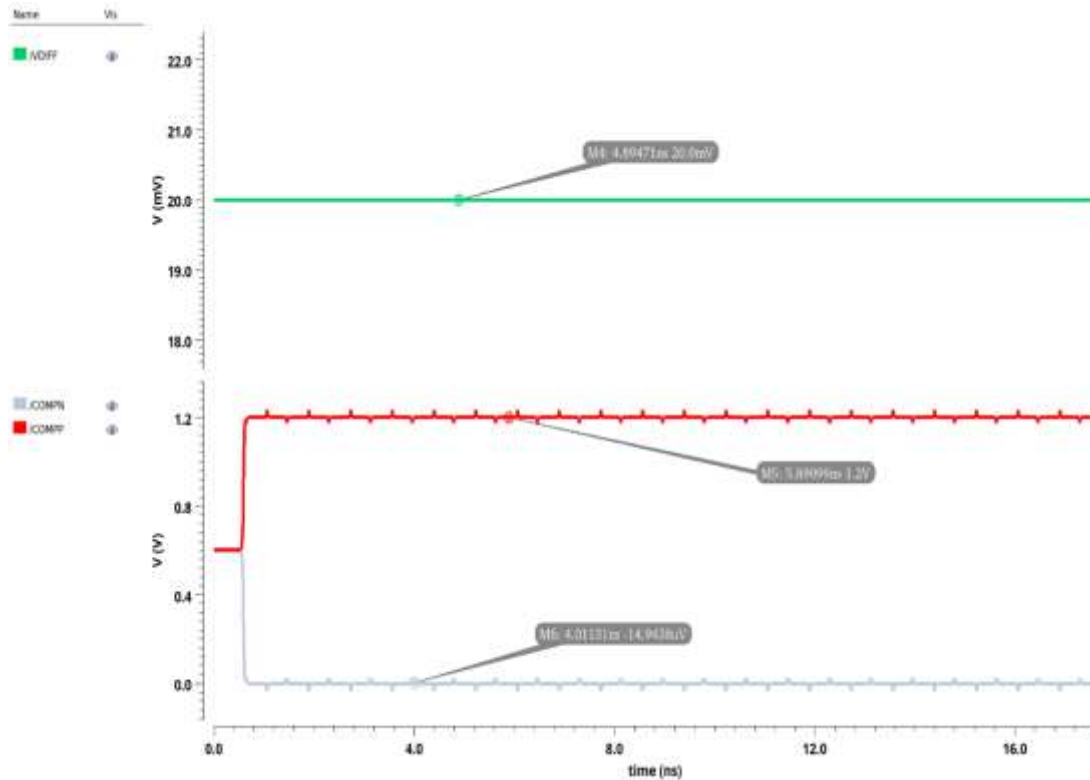


Figure 29: Comparator, buffer and SR latch output

2. Negative DC input to the comparator:

As shown in Fig 30, when a dc input (-20mV) as a differential input to the comparator, as this means that V_{INN} which is the V_{GS} of the NMOS is greater than V_{INP} which is the V_{GS} of the PMOS, so the output of the differential pair V_{OP} will discharge faster than V_{ON} , and this is obvious in fig 30, $V_{ON} = 421\text{mV} < V_{OP} = 405\text{mV}$. Then after regeneration starts, the output of the comparator V_{ON} goes to VDD and V_{OP} to GND. Then after the regeneration starts, the output of the comparator V_{ON} goes to VDD and V_{OP} to GND.

As shown in Fig 31, the output of the buffer is smoother than the output of the comparator, V_{ON} (buffered V_{ON} negative output of the comparator) = VDD and V_{OP} (buffered V_{OP} positive output of the comparator) = GND.

As shown in Fig 32, the output of SR latch insensitive to what happens in the reset phase, $OUTN$ (will be negative output of the comparator block in the system) = VDD and $OUTP$ (will be positive output of the comparator block in the system) = GND.

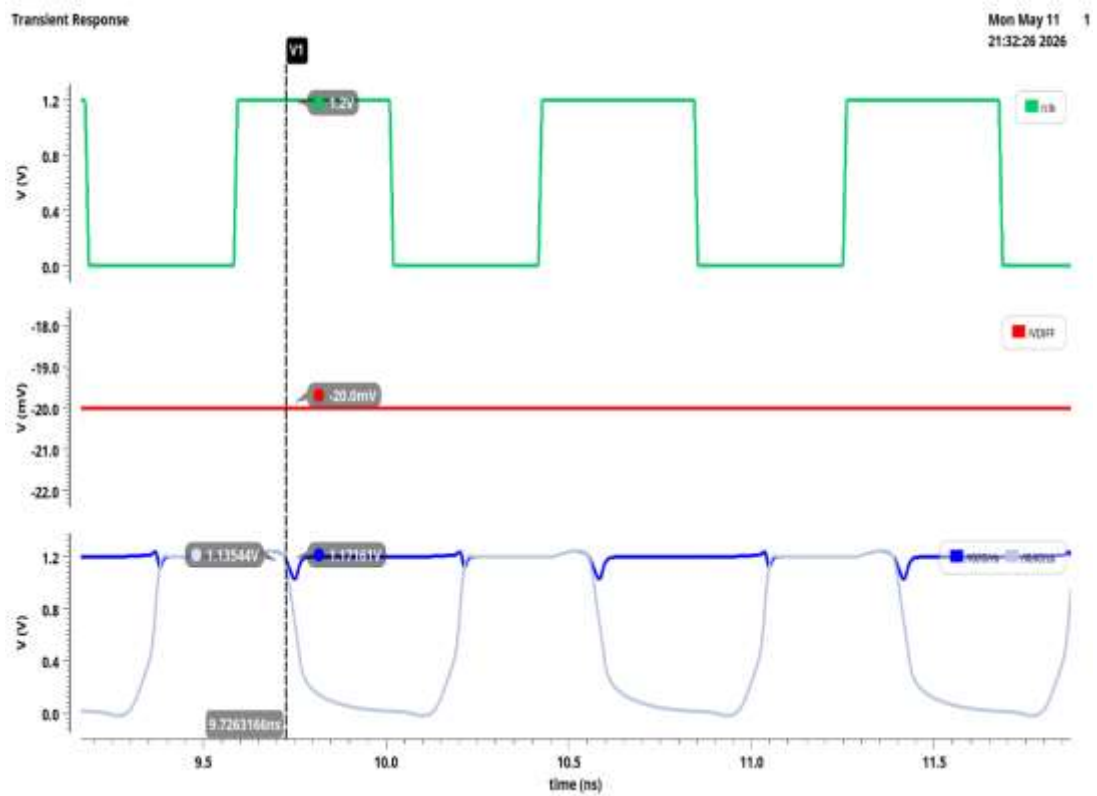


Figure 30: diff pair output and comparator output

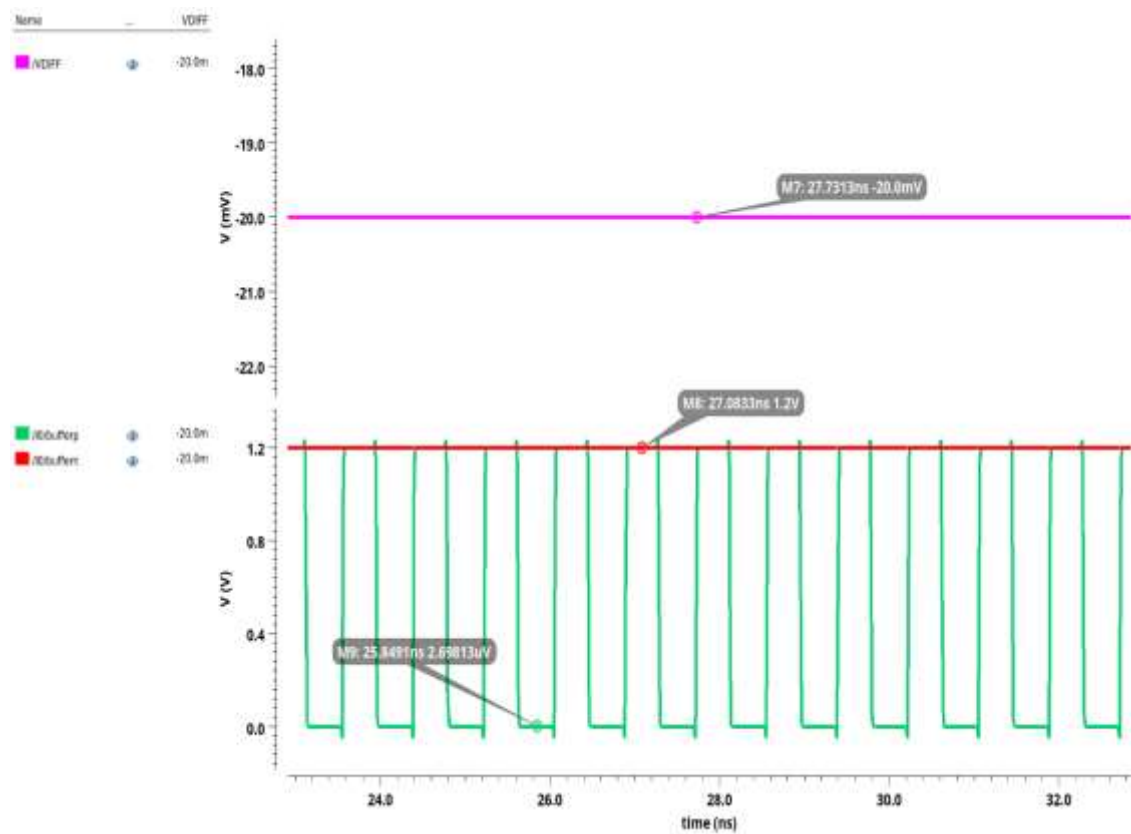


Figure 31: Comparator and buffer output

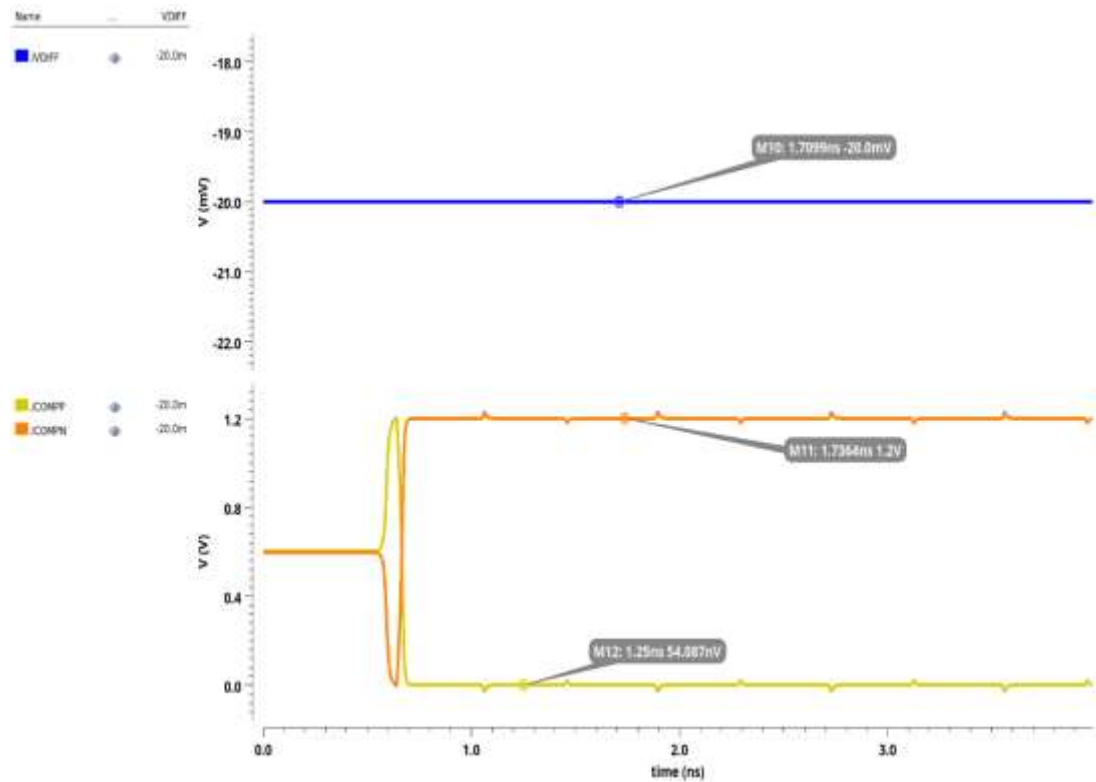


Figure 32: Comparator, buffer and SR latch output

3. Delay of the comparator block (after SR Latch):

As shown in Fig 33, the delay equal 169 pSec which is relatively not too large and not violate the required spec.

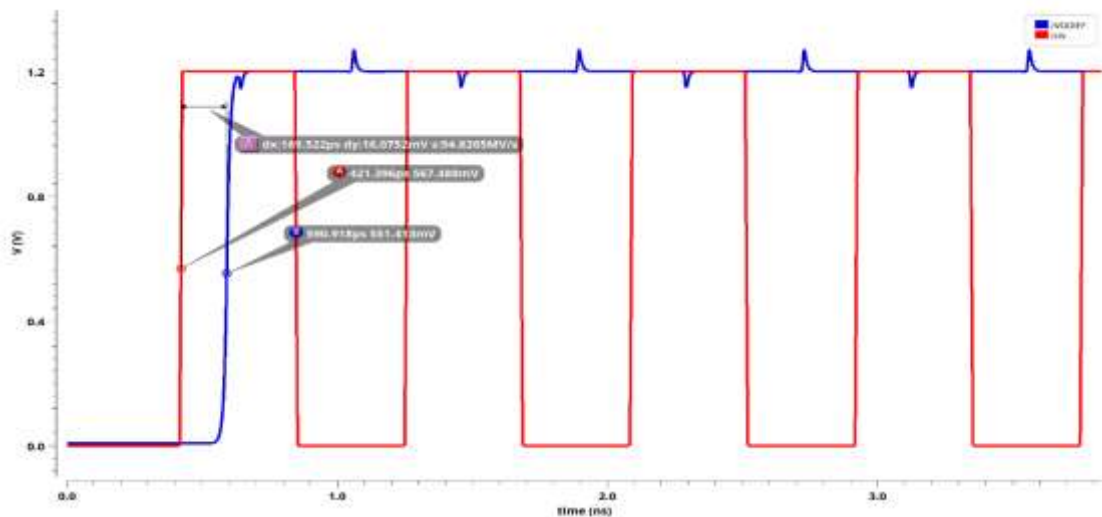


Figure 33: Delay of the Strong Arm used after SR latch

4. Power of the comparator:

Power of the comparator equals to 288.4 uw.

This test is done to check that the comparator has no hysteresis (the previous decision doesn't affect the next decision), so we have to put a full-scale input followed by $\pm 0.5\text{LSB}$.

As shown in Fig 34, we put full scale input (1.2 V) and then at the next cycle the input $< +0.5\text{LSB} = 937\mu\text{v}$. It's obvious that VOP of the comparator = VDD in the cycle at which the input was 1.2, VON = GND. And VOP of the comparator = VDD in the cycle at which the input was 937 μv .



As shown in Fig 35, we put full scale input (1.2 V) and then at the next cycle the input $< -0.5\text{LSB} = -937\mu\text{v}$. It's obvious that VOP of the comparator = VDD in the cycle at which the input was 1.2, VON = GND. And VON of the comparator = VDD in the cycle at which the input was -937 μv , VOP = GND.

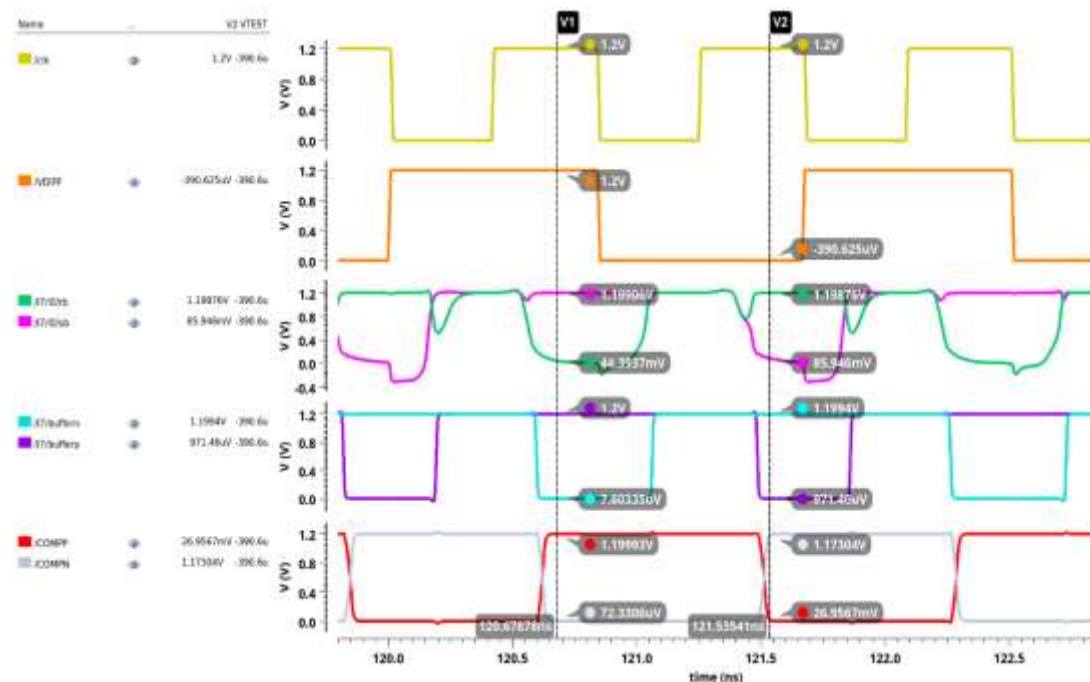


Figure 35: Negative over drive recovery test (1.2V, -2m V input)

6. Corners Simulations:

In this section we will simulate 3 cases of corners (C0, C1 and C3). But we will test these corners, while doing negative over derive test for hysteresis, as it is more general case including many tests (negative small differential input, positive large differential input, positive large differential input followed by negative small differential input).

- **First corner temperature= -30 °K, supply voltage = 1,1.2,1.4v**
As shown in fig 36, we will find that in 4 corners (supply 1V, 1.2V, 1.4V) in (fs, sf, ss, ff) corners and outputs of SR latch is wrong as NMOS in slow case doesn't have time enough to discharge outputs to GND or PMOS in slow case doesn't have time enough to charge outputs to VDD.
- **Second corner temperature= 27 °K, supply voltage = 1,1.2,1.4v**
As shown in Fig 37, we will find that in 4 corners (supply 1V, 1.2V, 1.4V) in (fs, sf, ss, ff) corners and outputs of SR latch is wrong as NMOS in slow case doesn't have time enough to discharge outputs to GND or PMOS in slow case doesn't have time enough to charge outputs to VDD. And to enhance the comparator to improve this response, i tried to do different combinations in sizing the PMOSs and NMOSs of comparator. But it's too complex to size the transistors to solve all issues in corners and remain the sizes of the differential pair as it is to not violate the spec on the input capacitance.
- **Third corner temperature= 150 °K, supply voltage = 1,1.2,1.4v**
As shown in fig 38, we will find that in 4 corners (supply 1V, 1.2V, 1.4V) in (fs, sf, ss, ff) corners and outputs of SR latch is wrong as NMOS in slow case doesn't have time enough to discharge outputs to GND or PMOS in slow case doesn't have time enough to charge outputs to VDD.

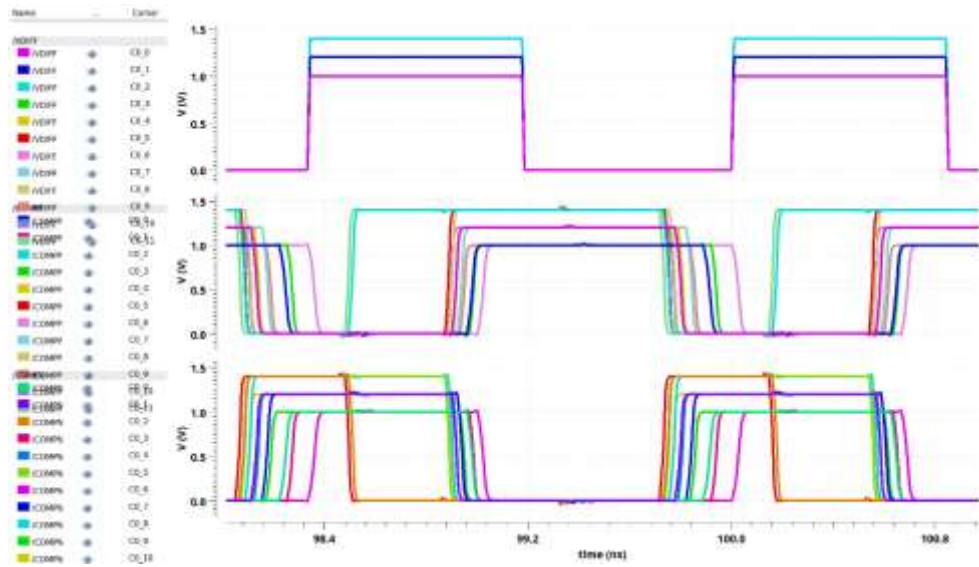


Figure 36: corners with temperature -30 °K, supply voltage 1, 1.2, 1.4 V

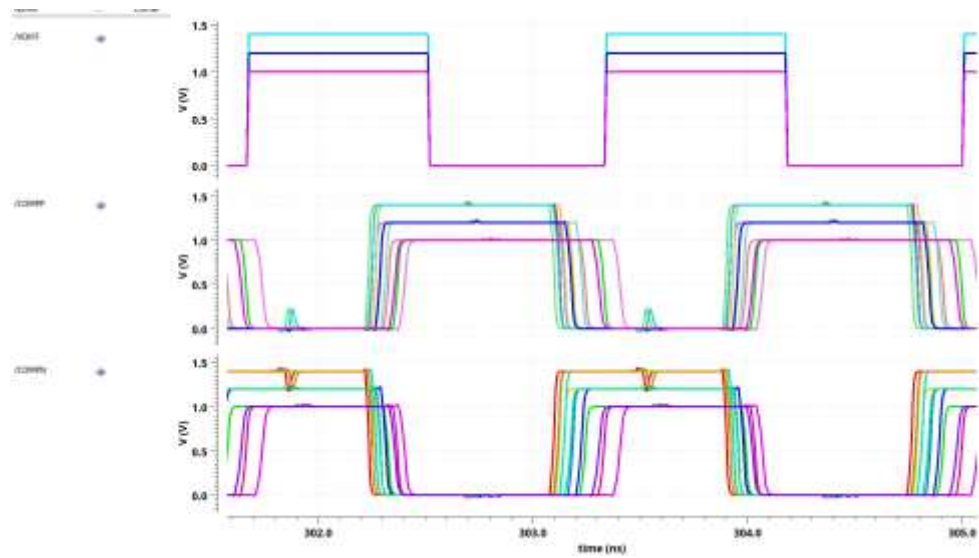


Figure 37: corners with temperature 27 °K, supply voltage 1, 1.2, 1.4 V

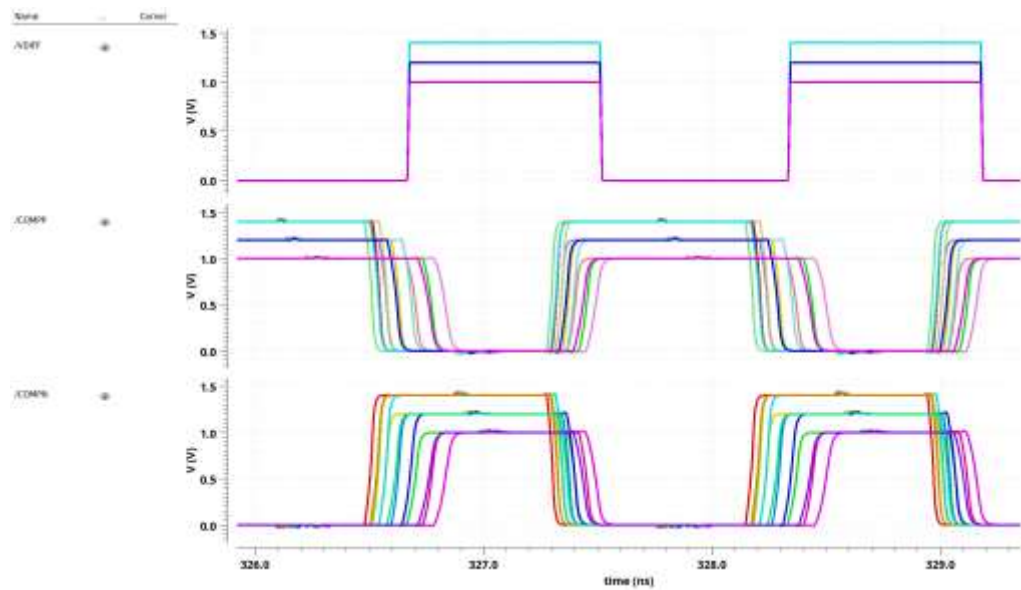


Figure 38: corners with temperature 150 °K, supply voltage 1, 1.2, 1.4 V

7. Offset Simulation:

In this section we will test the offset voltage, it is difficult to measure the offset voltage of a dynamic comparator since it does not have a quiescent steady state. To measure the offset voltage, you will apply a voltage staircase to the input of the dynamic comparator and latch the comparator at each step of the ramp. The input voltage level that causes the output to change is recorded. Then, the ramp direction is reversed, and the process is repeated. The offset voltage is the average of the two measurements. A slow ramp, staircase, is applied as the input of the comparator, and the comparator is latched for step of the staircase. You will record the staircase value when the comparator trips. The offset voltage is measured twice (once for the ramp up and once for the ramp down) and the result is averaged. Figure 39 illustrates both the input and latch out waveforms.

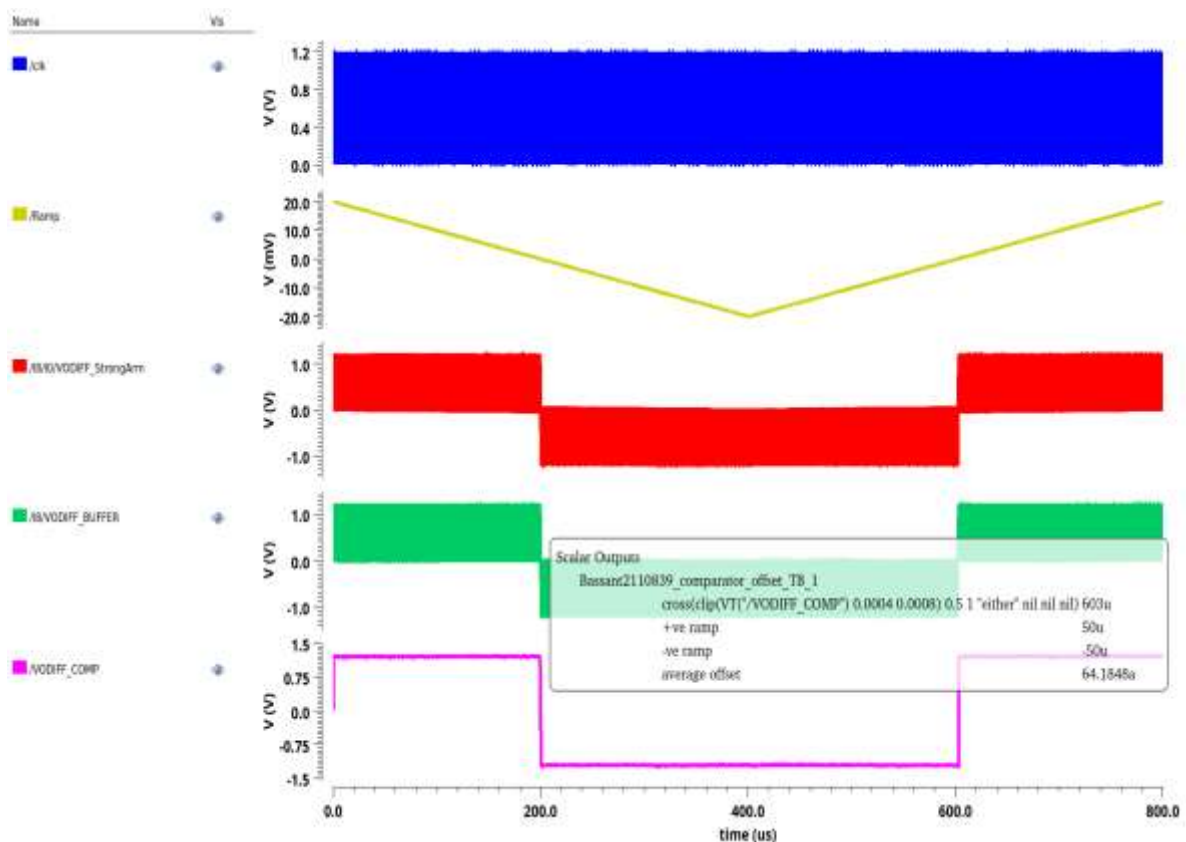


Figure 39: Comparator offset voltage

▪ Monte Carlo simulations for V_{OS}

To characterize the offset voltage of the comparator, the effect of global, wafer-to-wafer process variation and local, device-to-device process variation needs to be considered. You will use ADE Assembler to perform Monte Carlo simulation of the circuit.

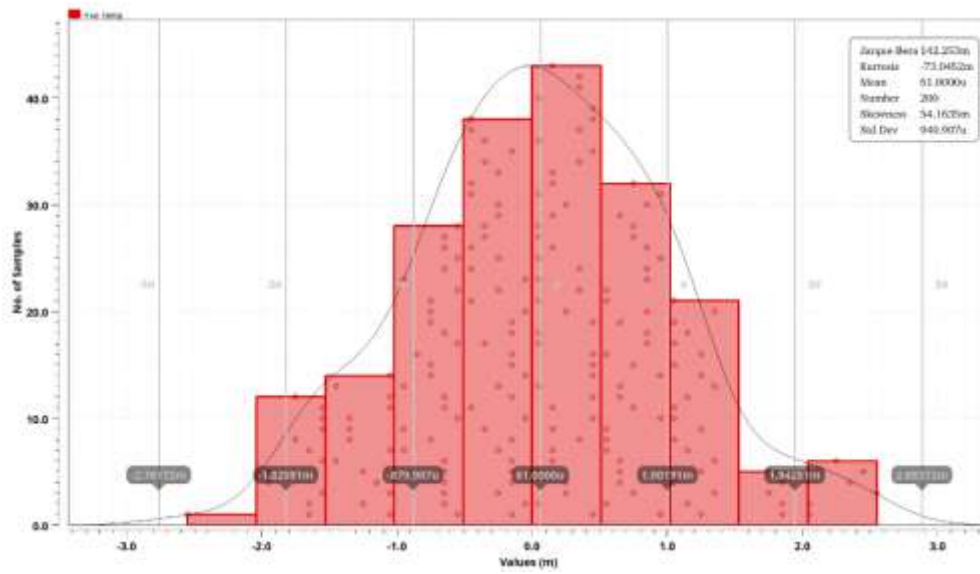


Figure 40: Monte Carlo simulations for VOS +

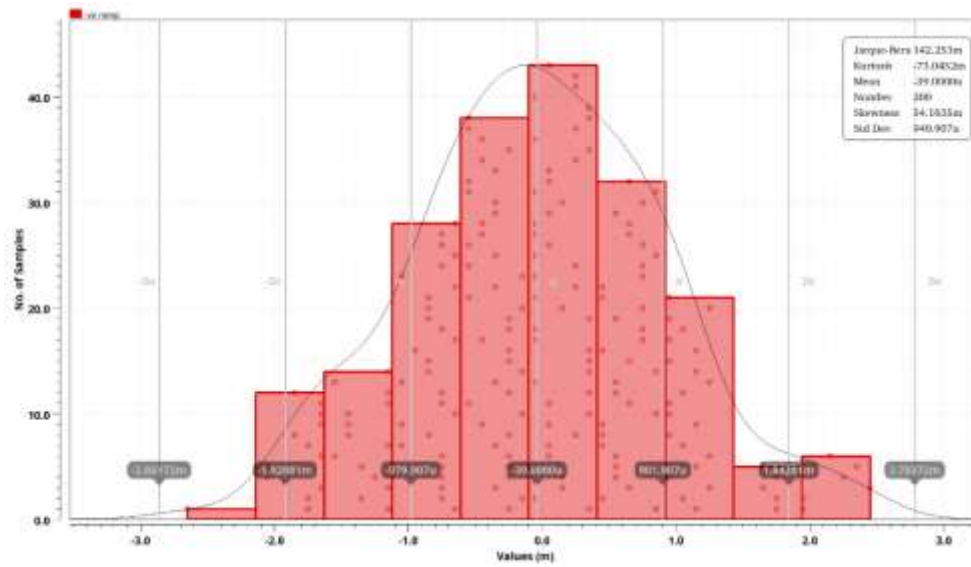


Figure 41: Monte Carlo simulations for VOS-

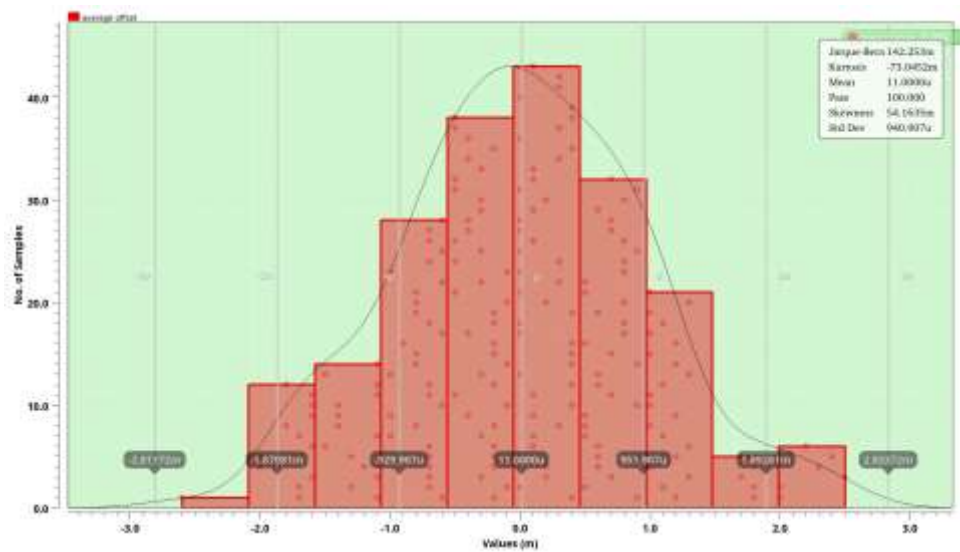


Figure 42: Monte Carlo simulations for average VOS

8. Input Capacitance Simulation:

Table 5.2: Input capacitance

NODE	Capacitance
VINP	102.2f
VINN	102.2f

Chapter 6

SAR Logic

6.1 SAR Logic Operation

Successive approximation register ADC implements the binary search algorithm using SAR control logic to change the voltage on the capacitors of the CDAC to determine the digital output signals (D9 to D0). SAR control logic determines the value of bits sequentially based on the result of the comparator and switching scheme.

SAR logic operation can be represented using a sequential Finite State Machine or with clock cycle diagram as shown in Fig 43. Each conversion takes 12 clock cycles. In the first clock cycle, SAR is in the reset mode and all the outputs are zero. In the next 10 clock cycles, data is converted and each bit is determined sequentially. The last cycle is for storing the results of the complete conversion.

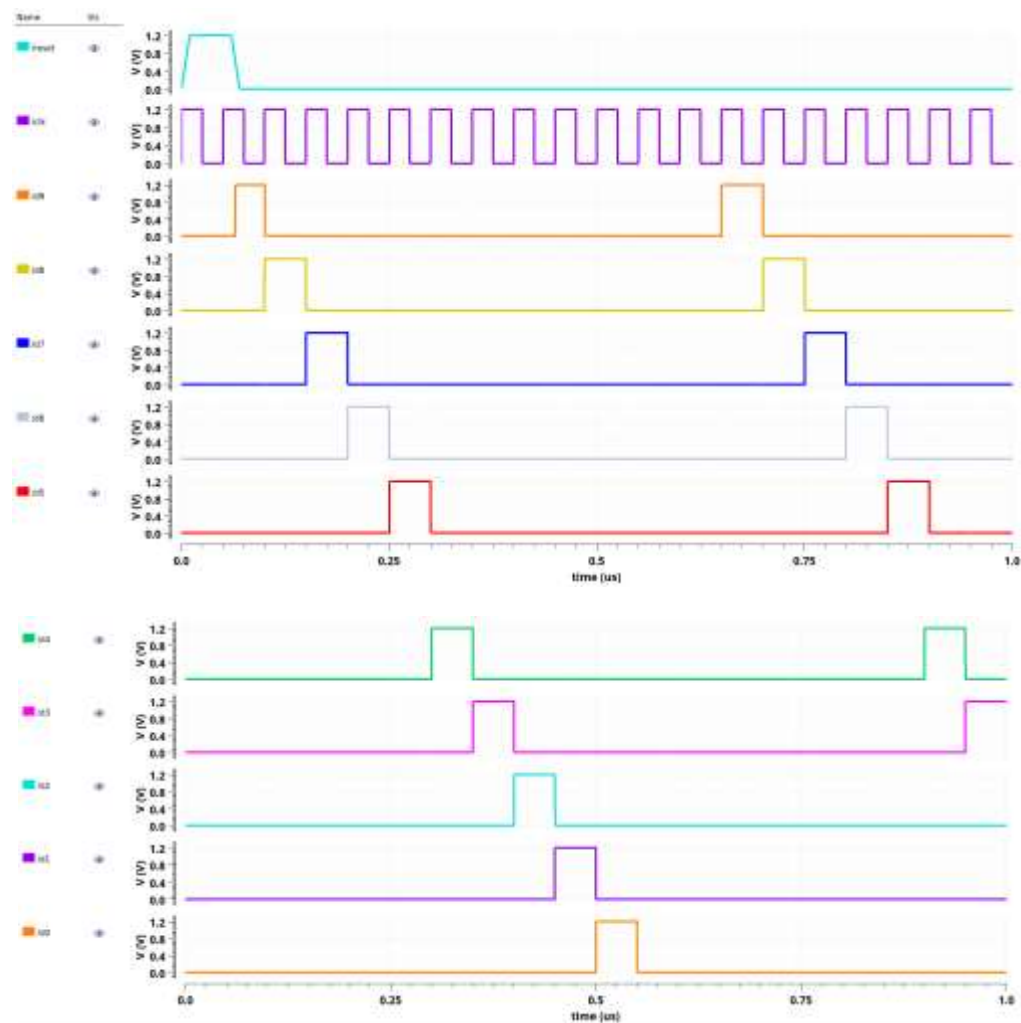


Figure 43: SAR logic time diagram

6.2 SAR Logic Topologies

The finite state machine algorithm can be separated to two types of SAR logic implemented topologies which will be discussed in this section.

6.2.1 SAR logic type 1

SAR logic architecture shown in Fig 44 is commonly used in SAR ADCs due to its straightforward design technique. This control logic contains a ring counter and a code register. The ring counter is in fact a shift register.

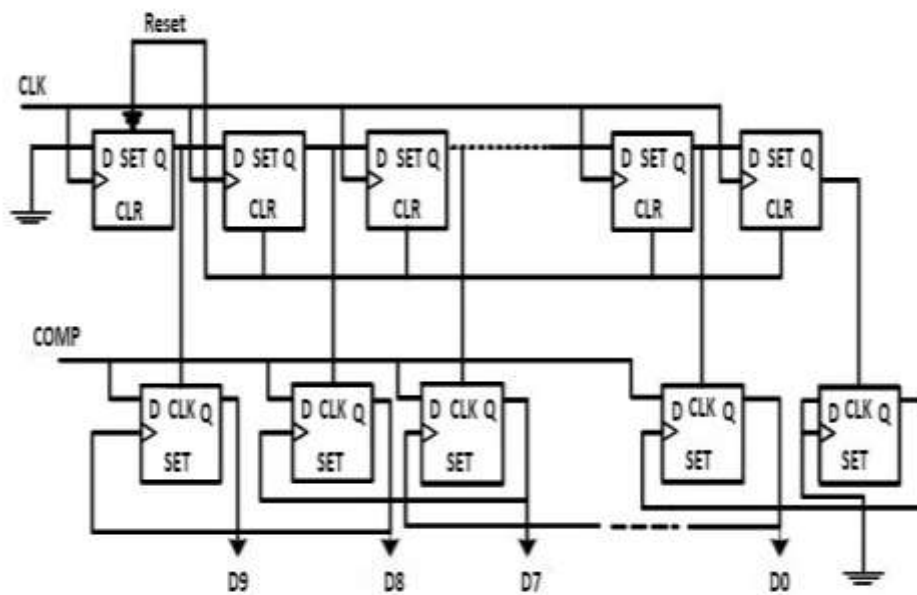


Figure 44: SAR logic type 1

For each conversion, in clock cycle 0, all Flip Flops outputs are reset in the two registers to zero except the first D Flip-flop in the shift registers is set to start conversion after sampling clock. Then at the next clock the output of this Flip Flop which is set by the ring counter is used as the clock signal for the Flip Flop. At rising edge of the clock, this Flip Flop of the code register loads the result from the comparator and determine the output of the most significant Flip Flop is which corresponds to MSB of the digital word to the DAC. Then the counter shifts '1' through the Flip-flops from MSB to LSB. It is low power due to low signal transition. Furthermore, since this logic is iterative, it can be extended to higher resolutions by just extending the shift registers. But its disadvantage is the area as it needs $2N$ flip-flops, N refers to number of bit.

6.2.2 SAR logic type 2

For an N-bit SAR ADC, digital control logic needs at least N-flip-flops to perform conversion. Therefore, a non-redundant SAR employs minimum number of flip-flops. These flip-flops both guess and store the converted result.

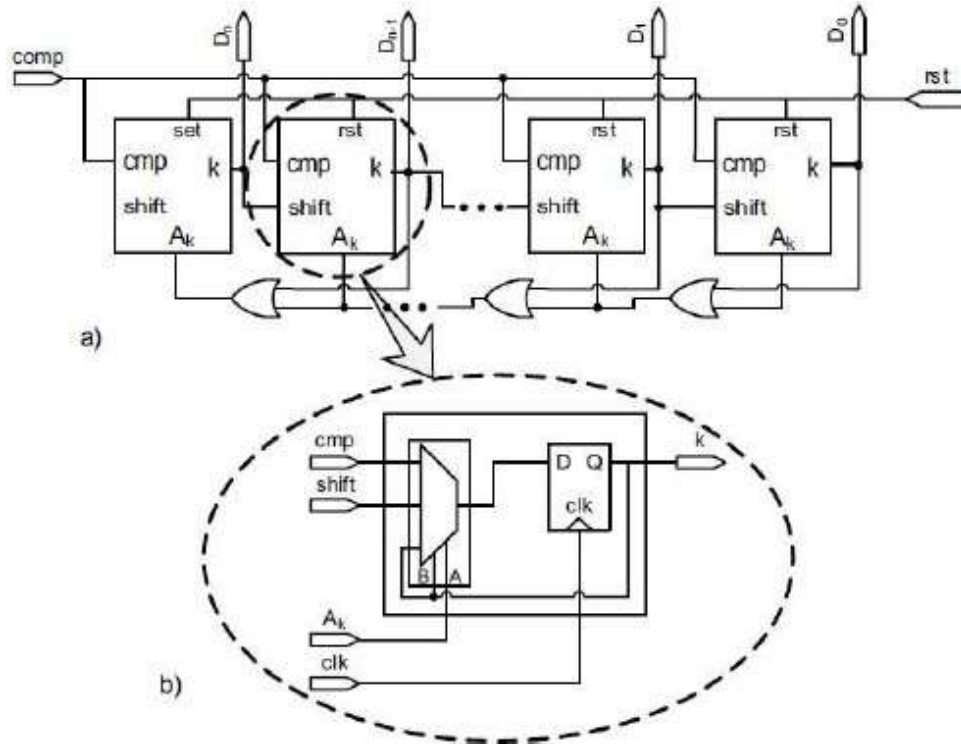


Figure 45: SAR logic type 2

As shown in Fig 45, In order to reduce the number of registers, a non-redundant SAR logic is proposed. For an N-bit SAR ADC converter, only N-registers are needed. The register has three functions: shift, input data, and storage. A and B are the control signals of the multiplexer. Start signal should be adjusted manually for different clock frequencies.

However, this problem is mitigated by adding a counter to the structure. Which is more hardware complicity and more power consumption, however its area is less than SAR logic type 1.

6.3 Decided Topology

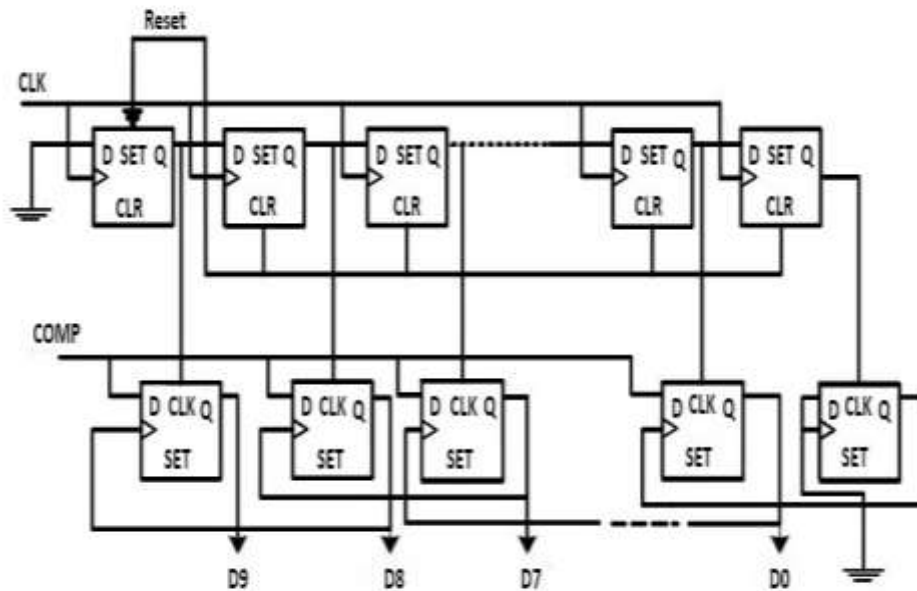
I used the SAR logic type 1 which is the ring register and the code register as I consider about power consumption and it is straight forward design techniques with adding improvement to achieve the required specification.

The specifications and requirements of our design is mentioned in table 6.1

Table 6.1: SAR logic specifications and requirements

Specifications	Values
Supply voltage	1.2V
Delay	< 200pSec
Rise and Fall time	10pSec
Power consumption	< 20 μ Watt

6.4 Circuit Implementation

*Figure 46: Circuit implementation*

6.4.1 Block implementation

The SAR logic consist of two main blocks as shown in the Fig 64 the shift registers and code registers, the two blocks consist of conventional D-Flip-Flop to form a register. The objective of the shift register is to be the clock signal for the Flip Flop of the code register which is used to determine the digital output. For each conversion, all the capacitors are reset to their original state either VDD or ground when the reset signal arrives. This reset signal is the sampling control signal. After reset, the logic is ready to sample the comparator output when the rising edge of the cycle control signal arrives. The sampling switch is the start switch for the SAR logic which is the reset signal to reset all the D-Flip-flops except the first D-Flip-flops to start the cycle from the beginning. After sampling switch resetting then the shift register starts to work by shifting the output '1' of the first D-Flip-flops through the Flip Flops from MSB to LSB. The code register works when the clock signal come from the shift register and decide the output based on loaded comparator result which will make the SAR control logic determines the value of bits sequentially.

The design of the D-Flip-flops will be discussed briefly in the next section.

6.4.2 D-Flip-flop implementation

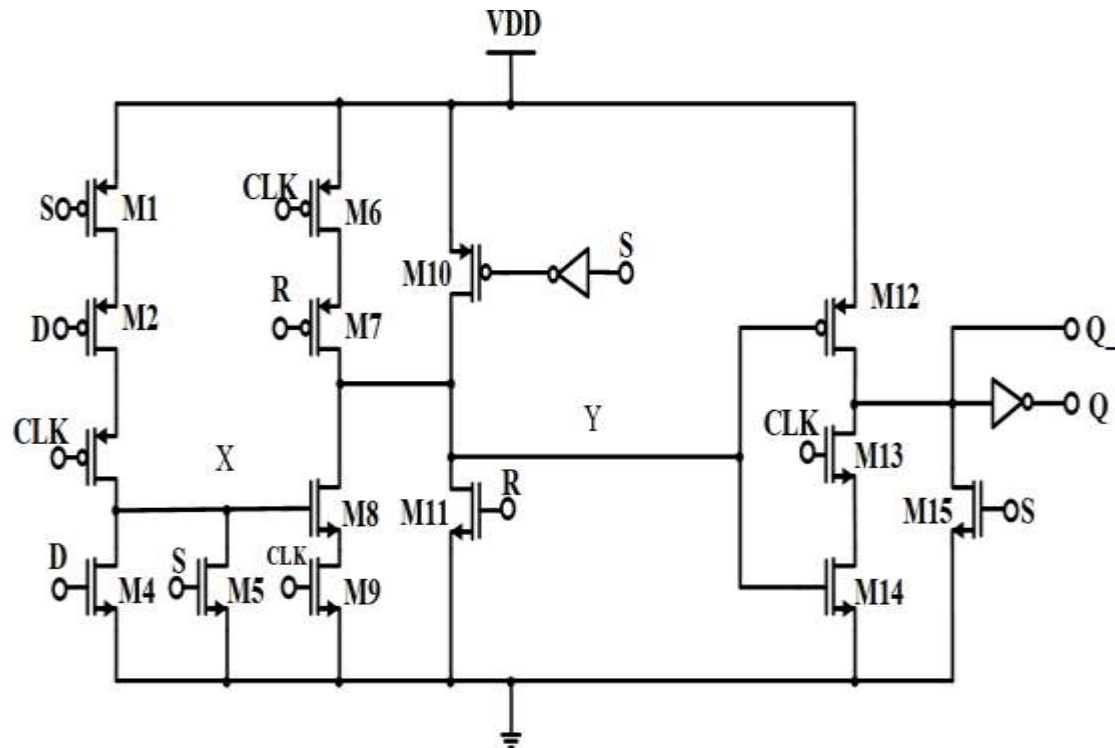


Figure 47: TSPC implementation

The D-Flip-flop of the shift register is True-single phase clock (TSPC) Flip-flops as it has the advantage of dynamic flip-flop, needs only a single clock signal, small area for clock lines which will lead to no clock skew and high speed. Using single phase clock and dynamic logic do not only lead to compact design but also faster response.

6.5 SAR Logic Testing and Results

6.5.1 Nominal Corner Simulations

Fig 48 shows that we achieved Delay time = 52.7 PS.

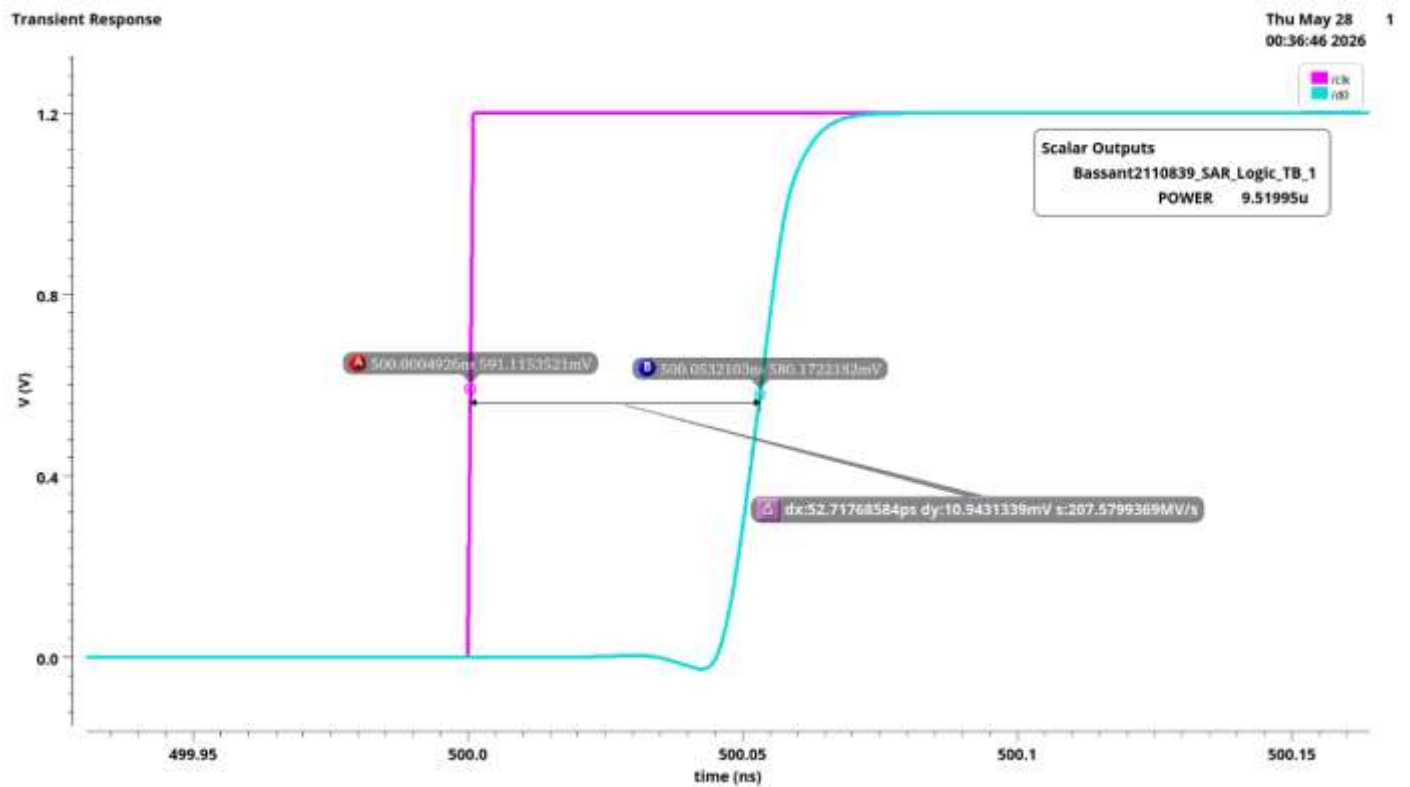


Figure 48: Delay analysis

6.5.2 Corner Simulations

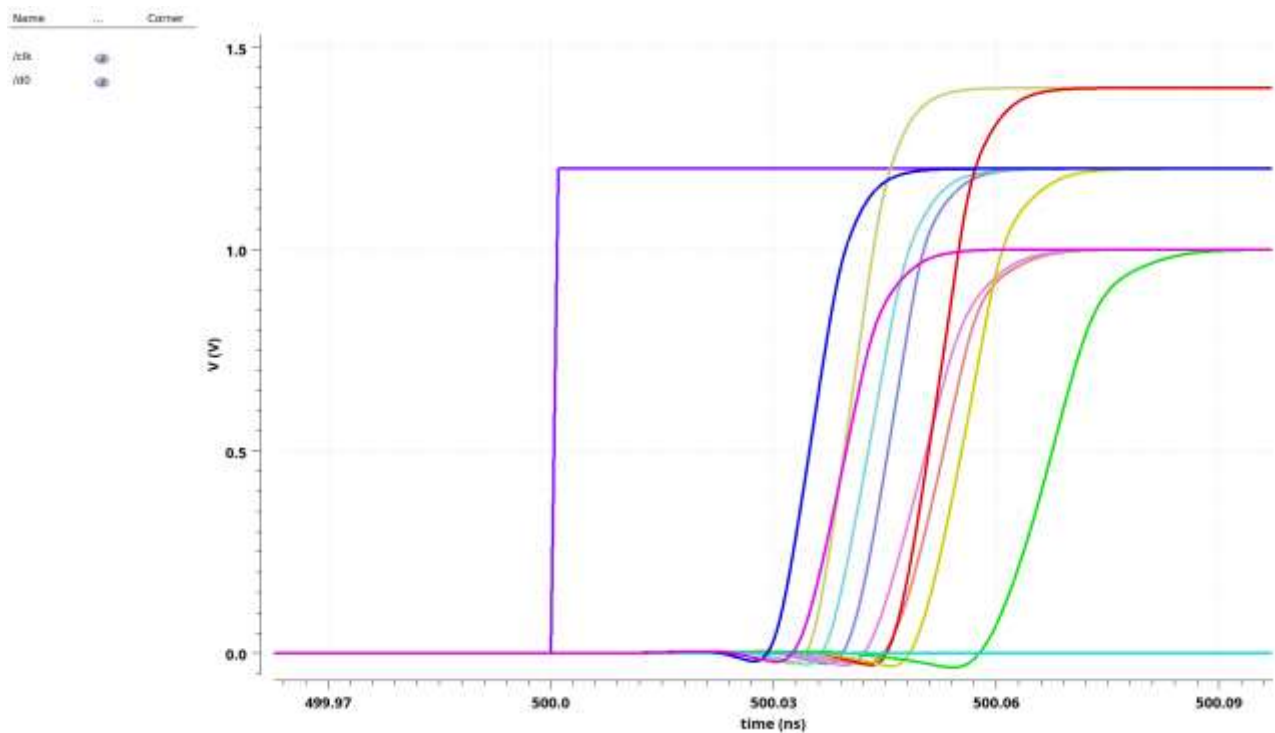


Figure 49: corners with temperature -30 °K, supply voltage 1, 1.2, 1.4 V

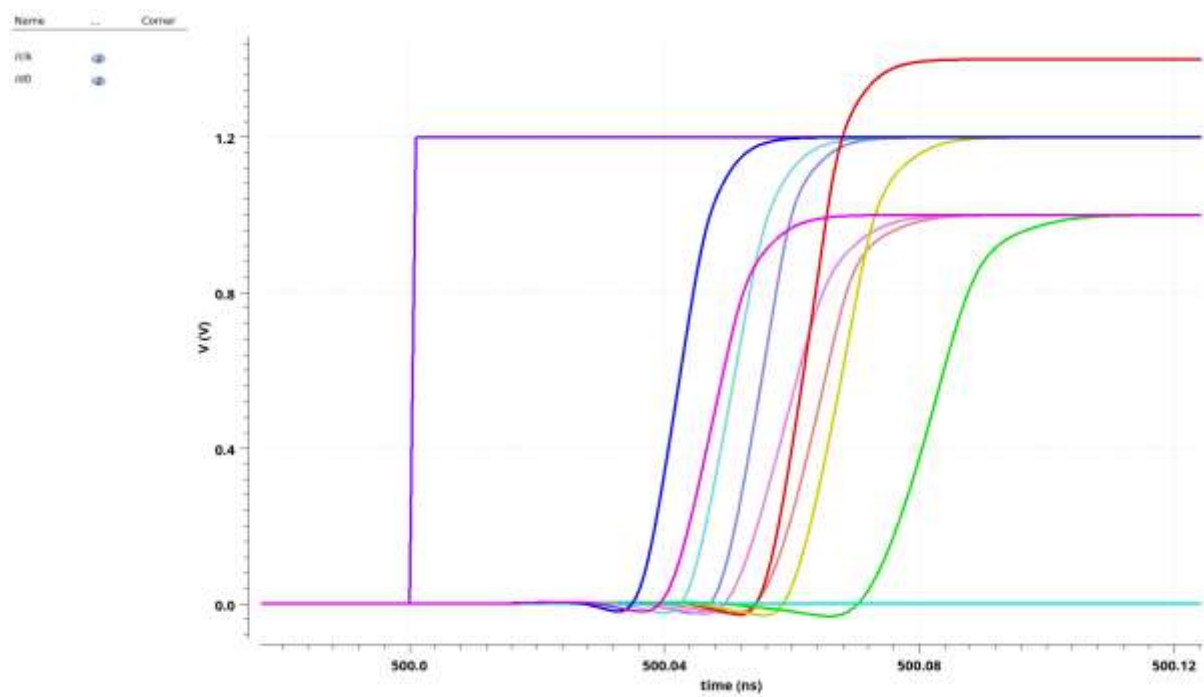


Figure 50: corners with temperature 27 °K, supply voltage 1, 1.2, 1.4 V

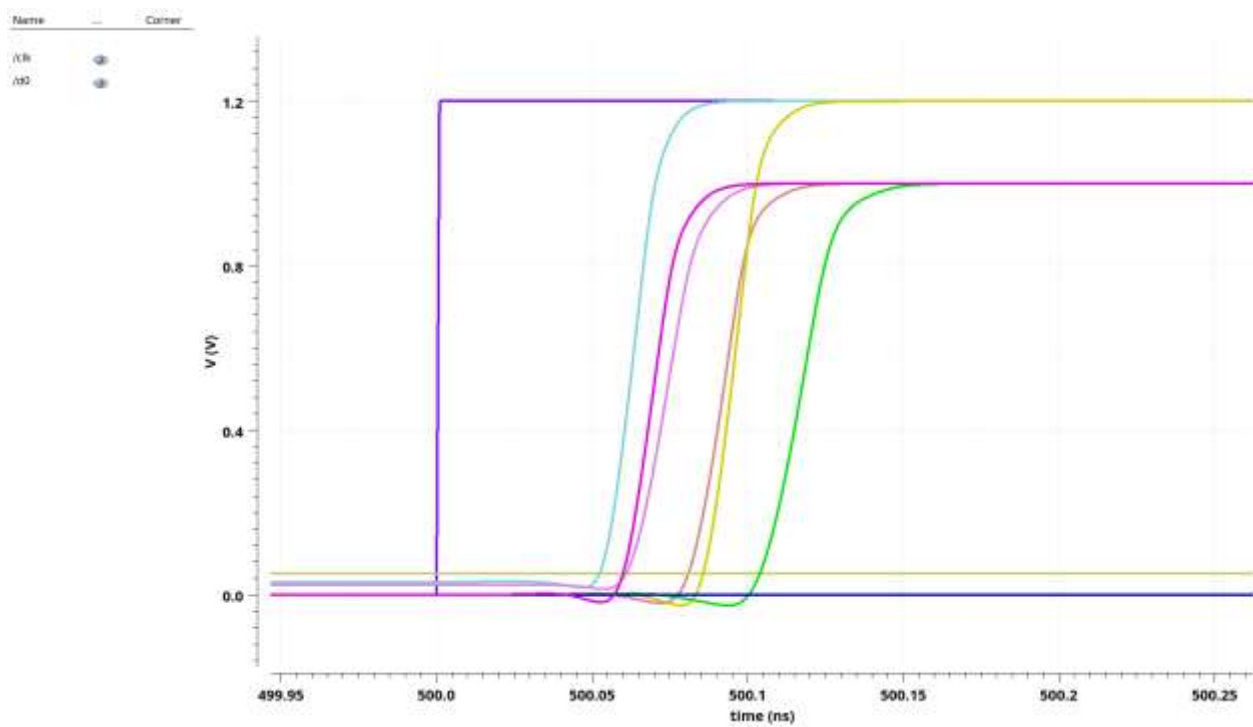


Figure 51: corners with temperature 150 °K, supply voltage 1, 1.2, 1.4 V

6.6 Achieved Specification

Table 6.2: SAR logic Achieved results

Specifications	Achieved
Supply voltage	-
Temperature	-
Delay	53ps
Rise and Fall time	10ps
Power consumption	9.5uw

Chapter 7

System Integration

This chapter discusses performance of the ADC in typical conditions. I will show the power consumption and distribution, Resolution for specific test input signal.

The test bench used for testing uses an input pure tone signal with value of frequency equals to 49.12 MHZ for nyquist operation.

7.1 Nyquist Operation

In this section we will show the results of the fully integrated system.

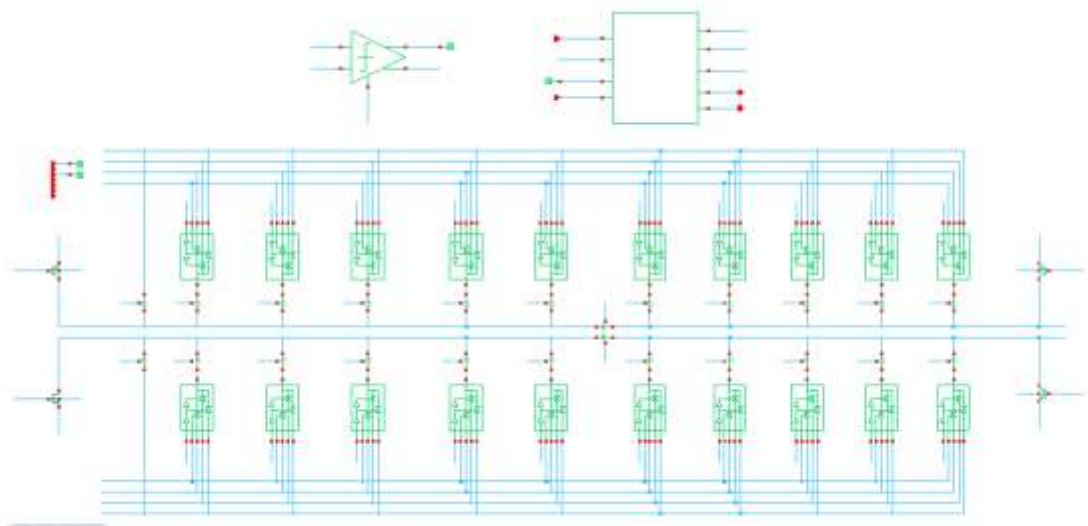


Figure 52: 10 Bit Asynchronous SAR ADC

7.2 Results

expr	[(v("Vout=0")-Result "tran") / 1024] + (v("Vout=9")-Res...			
expr	spectrumMeasurement((v("Vout=0")-Result "tran") / ...	9.954		
expr	spectrumMeasurement((v("Vout=0")-Result "tran") / ...	61.68		
expr	spectrumMeasurement((v("Vout=0")-Result "tran") / ...	61.89		
expr	spectrumMeasurement((v("Vout=0")-Result "tran") / ...	77.73		
expr	spectrumMeasurement((v("Vout=0")-Result "tran") / ...	-71.85		
signal	/vfp			
signal	/vfn			
signal	/vrefp			
signal	/vrefn			
signal	/vcin			
signal	/CLK			
signal	/rsc			
signal	/data_ready			
signal	/out<0:9>			
signal	/vout			
signal	/i0/outp			
signal	/i0/sample_B			
signal	/i0/bit<0:9>			
signal	/i0/sample			
signal	/i0/sample_clk			
signal	/i0/samp			
signal	/i0/summ			

Figure 53: ENOB of SAR ADC